Some elementary integrals

1. $\int 1\,dx = x + C$
2. $\int x\,dx = \frac{1}{2}x^2 + C$
3. $\int x^2\,dx = \frac{1}{3}x^3 + C$
4. $\int \frac{1}{x^2}\,dx = -\frac{1}{x} + C$
5. $\int \sqrt{x}\,dx = \frac{2}{3}x^{3/2} + C$
6. $\int \frac{1}{\sqrt{x}}\,dx = 2\sqrt{x} + C$
7. $\int x^r\,dx = \frac{1}{r+1}x^{r+1} + C \quad (r \neq -1)$
8. $\int \frac{1}{x}\,dx = \ln|x| + C$
9. $\int \sin ax\,dx = -\frac{1}{a}\cos ax + C$
10. $\int \cos ax\,dx = \frac{1}{a}\sin ax + C$
11. $\int \sec^2 ax\,dx = \frac{1}{a}\tan ax + C$
12. $\int \csc^2 ax\,dx = -\frac{1}{a}\cot ax + C$
13. $\int \sec ax\,\tan ax\,dx = \frac{1}{a}\sec ax + C$
14. $\int \csc ax\cot ax\,dx = -\frac{1}{a}\csc ax + C$
15. $\int \frac{1}{\sqrt{a^2 - x^2}}\,dx = \sin^{-1}\frac{x}{a} + C \quad (a > 0)$
16. $\int \frac{1}{a^2 + x^2}\,dx = \frac{1}{a}\tan^{-1}\frac{x}{a} + C$
17. $\int e^{ax}\,dx = \frac{1}{a}e^{ax} + C$
18. $\int b^{ax}\,dx = \frac{1}{a\ln b}b^{ax} + C$
19. $\int \cosh ax\,dx = \frac{1}{a}\sinh ax + C$
20. $\int \sinh ax\,dx = \frac{1}{a}\cosh ax + C$

1. (a) $\int (x^4 - 3x^3 + 8x^2 - 6x - 7)\,dx = \frac{x^5}{5} - \frac{3x^4}{4} + \frac{8x^3}{3} - 3x^2 - 7x + C$

(b) $\int \left(5x^{3/5} - \frac{3}{2 + x^2}\right)dx = \frac{25}{8}x^{8/5} - \frac{3}{\sqrt{2}}\tan^{-1}\frac{x}{\sqrt{2}} + C$

(c) $\int (4\cos 5x - 5\sin 3x)\,dx = \frac{4}{5}\sin 5x + \frac{5}{3}\cos 3x + C$

2.
$$\int \frac{(x+1)^3}{x}\,dx = \int \frac{x^3 + 3x^2 + 3x + 1}{x}\,dx$$
$$= \int \left(x^2 + 3x + 3 + \frac{1}{x}\right)dx$$
$$= \frac{1}{3}x^3 + \frac{3}{2}x^2 + 3x + \ln|x| + C.$$

3. $$\int \frac{5x^2 + 2x - 3}{x + 2}\,dx = \int \left(5x - 8 + \frac{13}{x + 2}\right)dx = \frac{5}{2}x^2 - 8x + 13\ln|x + 2| + C.$$

4.
$$\int \left(x - 2x^2 + \frac{1}{3x}\right)dx = \int \left(x - 2x^2 + \frac{1}{3}\cdot\frac{1}{x}\right)dx$$
$$= \int x\,dx - 2\int x^2\,dx + \frac{1}{3}\int \frac{1}{x}\,dx$$
$$= \frac{1}{2}x^2 - \frac{2}{3}x^3 + \frac{1}{3}\ln|x| + C$$

5. $$\int\left(\frac{7}{2x^3}-\sqrt[3]{x}\right)dx=\int\left(\frac{7}{2}x^{-3}-x^{1/3}\right)dx$$
$$=\frac{7}{2}\int x^{-3}dx-\int x^{1/3}dx$$
$$=\frac{7}{2}\left(-\frac{1}{2}\right)x^{-2}-\frac{1}{\frac{1}{3}+1}x^{4/3}+C$$
$$=-\frac{7}{4}x^{-2}-\frac{3}{4}x^{4/3}+C$$

6. $$\int -2\left(e^{2x}+1\right)dx=-2\int e^{2x}dx-2\int 1dx$$
$$=-2\left(\frac{1}{2}e^{2x}\right)-2x+C$$
$$=-e^{2x}-2x+C$$

7. $$\int\left(\frac{2}{\sqrt{x}}+2\sqrt{x}\right)dx=\int\left(2x^{-1/2}+2x^{1/2}\right)dx$$
$$=2\int x^{-1/2}dx+2\int x^{1/2}dx$$
$$=4x^{1/2}+\frac{4}{3}x^{3/2}+C$$
$$=4\sqrt{x}+\frac{4}{3}x^{3/2}+C$$

8. $$\int\left(8x^3-9x^2+4\right)dx=2x^4-3x^3+4x+C$$

9. $$\int\frac{x+6}{\sqrt{x}}\,dx=\int\left(x^{1/2}+6x^{-1/2}\right)dx$$
$$=\frac{x^{3/2}}{3/2}+6\frac{x^{1/2}}{1/2}+C$$
$$=\frac{2}{3}x^{3/2}+12x^{1/2}+C$$
$$=\frac{2}{3}x^{1/2}(x+18)+C$$

10. $$\int\frac{x^4-3x^2+5}{x^4}\,dx=\int\left(1-3x^{-2}+5x^{-4}\right)dx$$
$$=x-\frac{3x^{-1}}{-1}+\frac{5x^{-3}}{-3}+C$$
$$=x+\frac{3}{x}-\frac{5}{3x^3}+C$$

11. $$\int(5\cos x+4\sin x)\,dx=5\sin x-4\cos x+C$$

12. $$\int(1-\csc t\cot t)\,dt=t+\csc t+C$$

13. $\int \left(\sec^2\theta - \sin\theta\right) d\theta = \tan\theta + \cos\theta + C$

14. $$\int \sec y(\tan y - \sec y)\, dy = \int \left(\sec y \tan y - \sec^2 y\right) dy$$
$$= \sec y - \tan y + C$$

15. $$\int_{-1}^{1} e^{-t}\, dt = -e^{-t}\Big|_{-1}^{1} = -e^{-1} - (-e^{1}) = e - e^{-1} = e - \frac{1}{e} \approx 2.35.$$

16. $$\int \frac{3}{\sqrt{4-x^2}} + \cosh 4x\, dx = 3\int \frac{1}{\sqrt{2^2-x^2}} dx + \int \cosh 4x\, dx$$
$$= 3\sin^{-1}\left(\frac{x}{2}\right) + \frac{1}{4}\sinh 4x + c$$

17. $$\int \frac{1}{1+\cos x} dx = \int \frac{1}{1+\cos x} * \frac{1-\cos x}{1-\cos x} dx$$
$$= \int \frac{1-\cos x}{1-\cos^2 x} dx$$
$$= \int \frac{1-\cos x}{\sin^2 x} dx$$
$$= \int \frac{1}{\sin^2 x} - \frac{\cos x}{\sin^2 x} dx$$
$$= \int \csc^2 x - \frac{\cos x}{\sin x} * \frac{1}{\sin x} dx$$
$$= \int \csc^2 x - \cot x \csc x\, dx$$
$$= -\cot x + \csc x + c$$

18. $$\int (\sec x + \tan x)^2\, dx = \int \sec^2 x + \tan^2 x + 2\sec x \tan x\, dx$$
$$= \int 2\sec^2 x - 1 + 2\sec x \tan x\, dx$$
$$= 2\tan x - x + 2\sec x + c$$

19. $$\int \frac{2}{4x^2+9} dx = \frac{2}{4}\int \frac{1}{x^2+\frac{9}{4}} dx$$
$$= \frac{1}{2}\int \frac{1}{x^2+\left(\frac{3}{2}\right)^2} dx$$
$$= \frac{1}{2} * \frac{2}{3}\tan^{-1}\left(\frac{2}{3}x\right) + c$$

20. $$\int \csc x \cot x \sec x\, dx = \int \frac{1}{\sin x} \cdot \frac{\sin x}{\cos x} \cdot \frac{1}{\cos x} dx$$
$$= \int \frac{1}{\cos^2 x} dx$$
$$= \int \sec^2 x\, dx = \tan x + c$$

21.
$$\begin{aligned}\int \frac{(1+x)^2}{x^3+x}\,dx &= \int \frac{1+2x+x^2}{x\,(x^2+1)}\,dx \\ &= \int \frac{1+x^2}{x\,(x^2+1)} + \frac{2x}{x\,(x^2+1)}\,dx \\ &= \int \frac{1}{x} + \frac{2}{x^2+1}\,dx \\ &= \int \frac{1}{x}\,dx + 2\int \frac{1}{x^2+1}\,dx \\ &= \ln|x| + 2\tan^{-1}x + c\end{aligned}$$

the **method of substitution**, the integral version of the Chain Rule. If we rewrite the Chain Rule, $\frac{d}{dx} f\big(g(x)\big) = f'\big(g(x)\big)\,g'(x)$, in integral form, we obtain

$$\int f'\big(g(x)\big)\,g'(x)\,dx = f\big(g(x)\big) + C.$$

Let $u = g(x)$. Then $du/dx = g'(x)$, or in differential form, $du = g'(x)\,dx$. Thus,

$$\int f'\,(g(x))\;g'(x)\,dx = \int f'(u)\,du = f(u) + C = f\,(g(x)) + C.$$

22.
$$\int e^{5-2x}\,dx \qquad \begin{aligned}&\text{Let } u = 5-2x \\ &du = -2\,dx\end{aligned}$$
$$= -\frac{1}{2}\int e^u\,du = -\frac{1}{2}e^u + C = -\frac{1}{2}e^{5-2x} + C.$$

23.
$$\int \cos(ax+b)\,dx \qquad \begin{aligned}&\text{Let } u = ax+b \\ &du = a\,dx\end{aligned}$$
$$= \frac{1}{a}\int \cos u\,du = \frac{1}{a}\sin u + C$$
$$= \frac{1}{a}\sin(ax+b) + C.$$

24.
$$\int \sqrt{3x+4}\,dx \qquad \begin{aligned}&\text{Let } u = 3x+4 \\ &du = 3\,dx\end{aligned}$$
$$= \frac{1}{3}\int u^{1/2}\,du = \frac{2}{9}u^{3/2} + C = \frac{2}{9}(3x+4)^{3/2} + C.$$

25.
$$\int e^{2x}\sin(e^{2x})\,dx \qquad \begin{aligned}&\text{Let } u = e^{2x} \\ &du = 2e^{2x}\,dx\end{aligned}$$
$$= \frac{1}{2}\int \sin u\,du = -\frac{1}{2}\cos u + C$$
$$= -\frac{1}{2}\cos(e^{2x}) + C.$$

26. $\displaystyle\int \frac{x\,dx}{(4x^2+1)^5}$ Let $u = 4x^2 + 1$, $du = 8x\,dx$

$$= \frac{1}{8}\int u^{-5}\,du = -\frac{1}{32}u^{-4} + C = \frac{-1}{32(4x^2+1)^4} + C.$$

27. $\displaystyle\int \frac{\sin\sqrt{x}}{\sqrt{x}}\,dx$ Let $u = \sqrt{x}$, $du = \dfrac{dx}{2\sqrt{x}}$

$$= 2\int \sin u\,du = -2\cos u + C$$

$$= -2\cos\sqrt{x} + C.$$

28. $\displaystyle\int xe^{x^2}\,dx$ Let $u = x^2$, $du = 2x\,dx$

$$= \frac{1}{2}\int e^u\,du = \frac{1}{2}e^u + C = \frac{1}{2}e^{x^2} + C.$$

29. $\displaystyle\int \frac{\cos x}{4+\sin^2 x}\,dx$ Let $u = \sin x$, $du = \cos x\,dx$

$$= \int \frac{du}{4+u^2}$$

$$= \frac{1}{2}\tan^{-1}\frac{u}{2} + C = \frac{1}{2}\tan^{-1}\left(\frac{1}{2}\sin x\right) + C.$$

30. $\displaystyle\int \frac{\sec^2 x}{\sqrt{1-\tan^2 x}}\,dx$ Let $u = \tan x$, $du = \sec^2 x\,dx$

$$= \frac{du}{\sqrt{1-u^2}}$$

$$= \sin^{-1} u + C$$

$$= \sin^{-1}(\tan x) + C.$$

31. $\displaystyle\int \frac{\ln t}{t}\,dt$ Let $u = \ln t$, $du = \dfrac{dt}{t}$

$$= \int u\,du = \frac{1}{2}u^2 + C = \frac{1}{2}(\ln t)^2 + C.$$

32. $\displaystyle\int \tan x \ln\cos x\,dx$ Let $u = \ln\cos x$, $du = -\tan x\,dx$

$$= -\int u\,du = -\frac{1}{2}u^2 + C = -\frac{1}{2}\left(\ln\cos x\right)^2 + C.$$

33. $\displaystyle\int \frac{x+1}{\sqrt{x^2+2x+3}}\,dx$ Let $u = x^2+2x+3$, $du = 2(x+1)\,dx$

$$= \frac{1}{2}\int \frac{1}{\sqrt{u}}\,du = \sqrt{u} + C = \sqrt{x^2+2x+3} + C$$

34. $\displaystyle\int \frac{t}{\sqrt{4-t^4}}\,dt$ Let $u = t^2$, $du = 2t\,dt$

$$= \frac{1}{2}\int \frac{du}{\sqrt{4-u^2}}$$

$$= \frac{1}{2}\sin^{-1}\frac{u}{2} + C = \frac{1}{2}\sin^{-1}\left(\frac{t^2}{2}\right) + C.$$

35. $\int (x^2+7)^5(2x)\,dx \quad \Rightarrow \begin{Bmatrix} u = x^2+7 \\ du = 2x\,dx \end{Bmatrix}$

$$= \int u^5\,du$$

$$= \frac{u^6}{6} + C$$

$$= \frac{(x^2+7)^6}{6} + C$$

36. $\int \sin^9 x\,\cos x\,dx$

$$= \int (\sin x)^9 \cos x\,dx \quad \Rightarrow \begin{Bmatrix} u = \sin x \\ du = \cos x\,dx \end{Bmatrix}$$

$$= \int u^9\,du$$

$$= \frac{u^{10}}{10} + C$$

$$= \frac{\sin^{10} x}{10} + C$$

37. $\displaystyle\int (5x+4)^5\,dx$

(a) Let $u = 5x+4$

(b) Then $du = 5\,dx$ or $\frac{1}{5}\,du = dx$.

(c) Now substitute

$$\begin{aligned}\int (5x+4)^5\,dx &= \int u^5 \cdot \frac{1}{5}\,du \\ &= \int \frac{1}{5}u^5\,du \\ &= \frac{1}{30}u^6 + C \\ &= \frac{1}{30}(5x+4)^6 + C\end{aligned}$$

38. $\int \cos(2x+1)\,dx$

(a) Let $u = 2x+1$

(b) Then $du = 2\,dx$ or $\frac{1}{2}\,du = dx$

Now substitute

$$\begin{aligned}\int \cos(2x+1)\,dx &= \int \cos(u)\cdot\frac{1}{2}\,du \\ &= \int \frac{1}{2}\cos(u)\,du \\ &= \frac{1}{2}\sin(u)+C \\ &= \frac{1}{2}\sin(2x+1)+C\end{aligned}$$

39. $\int \sec^2(e^x)\cdot e^x dx \quad \Rightarrow \begin{Bmatrix} u = e^x \\ du = e^x dx \end{Bmatrix}$

$= \int \sec^2(u)\ du$

$= \tan u + C$

$= \tan(e^x) + C$

$\int e^{\sin x}\cdot \cos x\ dx \quad \Rightarrow \begin{Bmatrix} u = \sin x \\ du = \cos x\, dx \end{Bmatrix}$

$= \int e^u\ du = e^u + C$

$= e^{\sin x} + C$

From previous examples we can apply the following rule directly:

$$\int \underbrace{\big(f(x)\big)}_{\downarrow u}^{n}\cdot \underbrace{f'(x)\,dx}_{\downarrow du} = \frac{\big(f(x)\big)^{n+1}}{n+1}+C \qquad ,n \neq -1$$

40. $\int \frac{x^2-1}{(x^3-3x+2)^5}\,dx = \int (x^3-3x+2)^{-5}\cdot(x^2-1)\,dx$

$= \frac{1}{3}\int (x^3-3x+2)^{-5}\cdot 3(x^2-1)\,dx$

$= \frac{1}{3}\cdot\frac{(x^3-3x+2)^{-4}}{-4}+C$

$= \frac{-1}{12}(x^3-3x+2)^{-4}+C$

41. $\int (x^4-4x)\sqrt{x^5-10\,x^2+3}\ dx$

$= \frac{1}{5}\int 5(x^4-4x)(x^5-10\,x^2+3)^{\frac{1}{2}}\ dx$

$= \frac{1}{5}\cdot\frac{(x^5-10\,x^2+3)^{\frac{3}{2}}}{\frac{3}{2}}+C$

$= \frac{2}{15}(x^5-10\,x^2+3)^{\frac{3}{2}}+C$

42. $\int \tan^7(6x+3)\cdot\sec^2(6x+3)\,dx$

$=\frac{1}{6}\int \tan^7(6x+3)\cdot 6\sec^2(6x+3)\,dx$

$=\frac{1}{6}\cdot\frac{\tan^8(6x+3)}{8}+C$

$=\frac{1}{48}\tan^8(6x+3)+C$

$$\int (f(x))^n\cdot f'(x)\,dx$$

When $n=-1$

$\int (f(x))^{-1}\cdot f'(x)\,dx=\int\frac{f'(x)}{f(x)}\,dx\Rightarrow$

$\begin{Bmatrix} u=f(x) \\ du=f'(x)dx \end{Bmatrix}$

$=\int\frac{1}{u}\,du$

$=\ln|u|+C$

$=\ln|f(x)|+C$

$$\int\frac{f'(x)}{f(x)}\,dx=\ln|f(x)|+C$$

43. $\int\frac{2x+3}{x^2+3x-2}\,dx=\ln|x^2+3x-2|+C$

44. $\int\frac{x^2-1}{3x^3-9x}\,dx=\frac{1}{9}\int\frac{9(x^2-1)}{3x^3-9x}\,dx$

$=\frac{1}{9}\ln|3x^3-9x|+C$

45. $\int\frac{8}{x\ln x}\,dx=8\int\frac{1/x}{\ln x}\,dx=8\ln|\ln x|+C$

46. $\int\tan x\,dx=-\int\frac{-\sin x}{\cos x}dx=-\ln|\cos x|+C$

$=\ln|\sec x|+C$

47. $\int\cot x\,dx=\int\frac{\cos x}{\sin x}dx=\ln|\sin x|+C$

48. $\int\sec x\,dx=\int\sec x\frac{\sec x+\tan x}{\sec x+\tan x}dx$

$=\int\frac{\sec^2 x+\sec x\tan x}{\sec x+\tan x}dx=\ln|\sec x+\tan x|+C$

49. $\int \left(1+\frac{1}{t}\right)^3 \frac{1}{t^2}\,dt$

(a) Let $u = 1 + \frac{1}{t}$

(b) Then $du = -\frac{1}{t^2}\,dt$ or $-du = \frac{1}{t^2}\,dt$

(c) Now substitute

$$\begin{aligned}\int \left(1+\frac{1}{t}\right)^3 \cdot \frac{1}{t^2}\,dt &= \int u^3\,(-du)\\ &= -\frac{1}{4}u^4 + C\\ &= -\frac{1}{4}\left(1+\frac{1}{t}\right)^4 + C\end{aligned}$$

Using Substitution on Definite Integrals

50. Evaluate $\int_1^3 \frac{x}{\sqrt{2x-1}}dx.$

Solution:

Let $u = 2x - 1$.Then $du = 2dx$,or $dx = du/2$.Before we substitute, we need to determine the new limits of integration in terms of the $u$variable. To do so, we simply substitute the limits of integration into $u = 2x - 1$:

Lower limit: For $x = 1, u = 2(1) - 1 = 1.$

Upper limit: For $x = 3, u = 2(3) - 1 = 5.$

We now substitute $u$and the associated limits into the integral:

$$\int_1^5 \frac{x}{\sqrt{u}}\frac{du}{2}.$$

As you may notice, the variable $x$is still hanging there. To write it in terms of u,since $u = 2x - 1$,solving for x,we get, $x = (u + 1)/(2)$.Substituting back into the integral,

$$\begin{aligned}&= \int_1^5 \frac{u+1}{2\sqrt{u}}\frac{du}{2}\\ &= \frac{1}{4}\int_1^5 \frac{u+1}{\sqrt{u}}du\\ &= \frac{1}{4}\int_1^5 (u+1)u^{-1/2}du\\ &= \frac{1}{4}\int_1^5 (u^{1/2}+u^{-1/2})du\\ &= \frac{1}{4}\left|\frac{2u^{3/2}}{3}+\frac{2u^{1/2}}{1}\right|_1^5.\\ &= \frac{1}{4}\left(\left|\frac{2(5)^{3/2}}{3}+\frac{2(5)^{1/2}}{1}\right|-\left|\frac{2(1)^{3/2}}{3}+\frac{2(1)^{1/2}}{1}\right|\right).\\ &= \frac{4\sqrt{5}-2}{3}.\end{aligned}$$

51. Evaluate $\int_0^{\pi/4} \tan x \sec^2 x dx$.

Solution:

Try $u = \tan x$.Then $du = \sec^2 x dx, dx = du/\sec^2 x.$

Lower limit: For $x = 0, u = \tan 0 = 0.$

Upper limit: For $x = \pi/4, u = \tan \frac{\pi}{4} = 1.$

Thus

$$\begin{aligned}\int_0^{\pi/4} \tan x \sec^2 x dx &= \int_0^1 u du \\ &= \left|\frac{u^2}{2}\right|_0^1 \\ &= \frac{1}{2} - 0 = \frac{1}{2}.\end{aligned}$$

52. $\int_0^4 x^3(x^2+1)^{-1/2}\,dx$ Let $u = x^2 + 1, \quad x^2 = u - 1$

$du = 2x\,dx$

$$= \frac{1}{2}\int_1^{17} (u-1)u^{-1/2}\,du$$

$$= \frac{1}{2}\left(\frac{2}{3}u^{3/2} - 2u^{1/2}\right)\Big|_1^{17}$$

$$= \frac{17\sqrt{17}-1}{3} - (\sqrt{17}-1) = \frac{14\sqrt{17}}{3} + \frac{2}{3}.$$

Integration By Parts

Start with the product rule:

$$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$$

$$d(uv) = u\,dv + v\,du$$

$$d(uv) - v\,du = u\,dv$$

$$u\,dv = d(uv) - v\,du$$

$$\int u\,dv = \int (d(uv) - v\,du)$$

$$\int u\,dv = \int (d(uv)) - \int v\,du$$

$$\boxed{\int u\,dv = uv - \int v\,du}$$

This is the Integration by Parts formula.

$$\int u\,dv = uv - \int v\,du$$

u differentiates to zero (usually).

dv is easy to integrate.

The Integration by Parts formula is a "product rule" for integration.

53. $\int xe^x\,dx$

$\int u\,dv = uv - \int v\,du$

First, pick *u* and *dv*.

$u = x \qquad dv = e^x\,dx$

$= x\,e^x - \int e^x\,dx$

Now take the derivative of *u* and the integral of *dv*.

$du = dx \qquad v = e^x$

The second integral is much easier and more familiar than the original integral.

Use the formula for Integration by Parts.

$$= xe^x - e^x + C$$

When deciding what to choose for *u*, remember L I P E T.

L - logarithmic function
I - inverse trig function
P - polynomial function
E - exponential function
T - trigonometry function

This is usually the preference order in which you would want to choose *u*.

54. $\int \ln x\,dx$

Remember L I P E T.

$u = \ln x \qquad dv = dx$

$du = \frac{dx}{x} \qquad v = x$

$\int u\,dv = uv - \int v\,du$

$= x \ln x - \int \frac{x}{x}\,dx$

$= x \ln x - \int dx$

$= x \ln x - x + C$

Easy right?

Now what if we have a *u* and a *dv* that NEVER runs out of derivatives or integrals?…

55. $\int x^3 \ln x\,dx = uv - \int v\,du$

$$= (\ln x)\frac{x^4}{4} - \int \frac{x^4}{4}\frac{1}{x}\,dx$$

$$= (\ln x)\frac{x^4}{4} - \frac{1}{4}\int x^3\,dx$$

$$= \frac{x^4}{4}(\ln x) - \frac{1}{4}\,\frac{x^4}{4} + C$$

$$= \frac{x^4}{4}(\ln x) - \frac{x^4}{16} + C$$

56. $\int x \sin x\,dx$

$$\begin{aligned} u &= x, & dv &= \sin x\,dx \\ du &= 1\,dx & v &= -\cos x \end{aligned}$$

$$\begin{aligned} \int \overbrace{x}^{u}\overbrace{\sin x\,dx}^{dv} &= \overbrace{x}^{u}\overbrace{(-\cos x)}^{v} - \int \overbrace{(-\cos x)}^{v}\overbrace{1\,dx}^{du} \\ &= -x\cos x - \int \cos x\,dx \\ &= -x\cos x - \sin x + c \end{aligned}$$

57. $\int x \tan^{-1} x\,dx$

$$\begin{aligned} u &= \tan^{-1} x & dv &= x\,dx \\ du &= \frac{1}{1+x^2}dx & v &= \frac{x^2}{2} \end{aligned}$$

$$\begin{aligned} \int x\tan^{-1} x\,dx &= \frac{x^2}{2}\tan^{-1} x - \frac{1}{2}\int \frac{x^2}{1+x^2}dx \\ &= \frac{x^2}{2}\tan^{-1} x - \frac{1}{2}\int \frac{x^2+1-1}{1+x^2}dx \\ &= \frac{x^2}{2}\tan^{-1} x - \frac{1}{2}\int 1 - \frac{1}{1+x^2}dx \\ &= \frac{x^2}{2}\tan^{-1} x - \frac{1}{2}\left(x - \tan^{-1} x\right) + c \end{aligned}$$

58. $\int \sin^{-1}x\ dx$

$u = \sin^{-1}x$	$dv = dx$
$du = \frac{1}{\sqrt{1-x^2}}\ dx$	$v = x$

$$\int \sin^{-1}x\ dx = x\sin^{-1}x - \int \frac{x}{\sqrt{1-x^2}}\ dx$$

$$= x\sin^{-1}x - \frac{1}{-2}\int -2x\,(1-x^2)^{\frac{-1}{2}}dx$$

$$= x\sin^{-1}x + \frac{1}{2}\frac{(1-x^2)^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$= x\sin^{-1}x + (1-x^2)^{\frac{1}{2}} + C$$

59. $\int x\tan^{-1} x\,dx$ Let $U = \tan^{-1} x$, $dV = x\,dx$.
Then $dU = dx/(1+x^2)$, $V = \frac{1}{2}x^2$.

$$= \frac{1}{2}x^2\tan^{-1} x - \frac{1}{2}\int \frac{x^2}{1+x^2}dx$$

$$= \frac{1}{2}x^2\tan^{-1} x - \frac{1}{2}\int \left(1 - \frac{1}{1+x^2}\right)dx$$

$$= \frac{1}{2}x^2\tan^{-1} x - \frac{1}{2}x + \frac{1}{2}\tan^{-1} x + C.$$

60. $\int x^2 \ln x \, dx.$

$$dv = x^2\,dx \quad \Rightarrow \quad v = \int x^2\,dx = \frac{x^3}{3}$$

$$u = \ln x \quad \Rightarrow \quad du = \frac{1}{x}\,dx$$

$$\int u\,dv = uv - \int v\,du$$

$$\int x^2 \ln x\,dx = \frac{x^3}{3}\ln x - \int \left(\frac{x^3}{3}\right)\left(\frac{1}{x}\right) dx$$

$$= \frac{x^3}{3}\ln x - \frac{1}{3}\int x^2\,dx$$

$$= \frac{x^3}{3}\ln x - \frac{x^3}{9} + C.$$

Tabular Method

61. $\int x^2 \sin 4x\,dx.$

Alternate Signs	*u and Its Derivatives*	*v′and Its Antiderivatives*
+	x^2	$\sin 4x$
−	$2x$	$-\frac{1}{4}\cos 4x$
+	2	$-\frac{1}{16}\sin 4x$
−	0	$\frac{1}{64}\cos 4x$

Differentiate until you obtain 0 as a derivative.

$$\int x^2 \sin 4x\,dx = -\frac{1}{4}x^2\cos 4x + \frac{1}{8}x\sin 4x + \frac{1}{32}\cos 4x + C.$$

62. $\int e^x \sin x\,dx$

$$u = \sin x,\ dv = e^x dx$$

$$du = \cos x dx,\ v = e^x$$

$$= e^x \sin x - \int e^x \cos x dx$$

$$u = \cos x,\ dv = e^x dx$$

$$du = -\sin x,\ v = e^x$$

$$= e^x \sin x - e^x \cos x + \int e^x(-\sin x)dx$$

$$= e^x \sin x - e^x \cos x - \int e^x \sin x dx$$

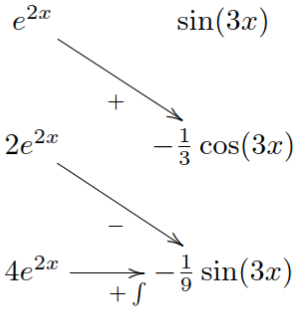
$$2\int e^x \sin x dx = e^x \sin x - e^x \cos x + C$$

$$\int e^x \sin x dx = \frac{1}{2}(e^x \sin x - e^x \cos x + C)$$

63.

$$\begin{aligned}
\int \sin^2 x dx \qquad & u = \sin x,\ dv = \sin x dx \\
& du = \cos x dx,\ v = -\cos x \\
= & -\sin x \cos x - \int (-\cos x)(\cos x) dx \\
= & -\sin x \cos x + \int \cos^2 x dx \\
= & -\sin x \cos x + \int (1 - \sin^2 x) dx \\
= & -\sin x \cos x + \int dx - \int \sin^2 x dx \\
= & -\sin x \cos x + x - \int \sin^2 x dx \\
2\int \sin^2 x dx = & -\sin x \cos x + x + C \\
\int \sin^2 x dx = & -\frac{1}{2}\sin x \cos x + \frac{1}{2}x + C
\end{aligned}$$

64. $\int e^{2x} \sin(3x) dx,$



$$\begin{aligned}
\int e^{2x}\sin(3)dx &= e^{2x}\cdot\left(-\frac{1}{3}\cos(3x)\right) - 2e^{2x}\cdot\left(-\frac{1}{9}\sin(3x)\right) + \int 4e^{2x}\cdot\left(-\frac{1}{9}\sin(3x)\right)dx \\
&= -\frac{1}{3}e^{2x}\cos(3x) + \frac{2}{9}e^{2x}\sin(3x) - \frac{4}{9}\int e^{2x}\sin(3x)dx
\end{aligned}$$

Tabular Method:

Example:

65. $\int x^3\, e^{2x}\, dx$

$= \frac{1}{2}x^3\, e^{2x} - \frac{3}{4}x^2\, e^{2x}$

$+\frac{6}{8}x\, e^{2x} - \frac{6}{16}\, e^{2x} + C$

+	x^3	e^{2x}
−	$3x^2$	$\frac{1}{2}e^{2x}$
+	$6x$	$\frac{1}{4}e^{2x}$
−	6	$\frac{1}{8}e^{2x}$
	0	$\frac{1}{16}e^{2x}$

66. $\int x^2 \cos 3x \ dx$

$= \frac{1}{3}x^2 \sin 3x + \frac{2}{9}x \cos 3x$

$-\frac{2}{27}\sin 3x + C$

+	x^2	$\cos 3x$
−	$2x$	$\frac{1}{3}\sin 3x$
+	2	$\frac{-1}{9}\cos 3x$
−	0	$\frac{-1}{27}\sin 3x$

67. $\int (x^3 + 2x) \sin 2x \ dx$

$= \frac{-1}{2}(x^3 + 2x)\cos 2x$

$+\frac{1}{4}(3x^2 + 2)\sin 2x$

$+\frac{6}{8}\ x \cos 2x$

$-\frac{6}{16}\sin 2x + C$

+	$x^3 + 2x$	$\sin 2x$
−	$3x^2+2$	$\frac{-1}{2}\cos 2x$
+	$6x$	$\frac{-1}{4}\ \sin 2x$
−	6	$\frac{1}{8}\cos 2x$
	0	$\frac{1}{16}\ \sin 2x$

68. $\int e^x \cos x \ dx$

$= e^x \sin x + e^x \cos x$

$-\int e^x \cos x \ dx$

+	e^x	$\cos x$
−	e^x	$\sin x$
$+\int$	e^x	$-\cos x$

$2\int e^x \cos x \ dx = e^x \sin x + e^x \cos x$

$\int e^x \cos x \ dx = \frac{1}{2}[e^x \sin x + e^x \cos x] + C$

69. $\int e^{3x} \cos 2x \ dx$

+	e^{3x}	$\cos 2x$
−	$3e^{3x}$	$\frac{1}{2}\sin 2x$
$+\int$	$9e^{3x}$	$\frac{-1}{4}\cos 2x$

$\int e^{3x} \cos 2x \ dx$

$= \frac{1}{2}e^{3x}\sin 2x + \frac{3}{4}e^{3x}\cos 2x - \frac{9}{4}\int e^{3x}\cos 2x \ dx$

$\frac{13}{4}\int e^{3x}\cos 2x \ dx = \frac{1}{2}e^{3x}\sin 2x + \frac{3}{4}e^{3x}\cos 2x$

$\int e^{3x}\cos 2x \, dx = \frac{4}{13}[\frac{1}{2}e^{3x}\sin 2x + \frac{3}{4}e^{3x}\cos 2x] + C$

Definite Integration

$$\int_a^b f(x)dx = F(x)\Big|_a^b = F(b) - F(a)\,,$$

70. $\int_1^2 x^3 dx = \frac{1}{3+1}x^{3+1}\Big|_1^2$

$= \frac{1}{4}x^4\Big|_1^2 = \frac{1}{4}\times 2^4 - \frac{1}{4}\times 1^1$

$= \frac{1}{4}\times 16 - \frac{1}{4} = 4 - \frac{1}{4} = \frac{15}{4}\,.$

71.
$$\begin{aligned}\int_{-4}^{-2} 6x^2 dx &= \frac{6}{2+1}x^{2+1}\Big|_{-4}^{-2} \\ &= \frac{6}{3}x^3\Big|_{-4}^{-2} = 2x^3\Big|_{-4}^{-2} \\ &= 2\times(-2)^3 - 2\times(-4)^3 = -16+128 = 112\,.\end{aligned}$$

72.
$$\begin{aligned}\int_{-1}^{2} x dx &= \frac{1}{1+1}x^{1+1}\Big|_{-1}^{2} = \frac{1}{2}x^2\Big|_{-1}^{2} \\ &= \frac{1}{2}\times(2)^2 - \frac{1}{2}\times(-1)^2 \\ &= \frac{1}{2}\times 4 - \frac{1}{2}\times(+1) \\ &= 2 - \frac{1}{2} = \frac{3}{2}\,.\end{aligned}$$

73.
$$\begin{aligned}\int_{0}^{\pi} \sin(x) dx &= -\cos(x)\Big|_{0}^{\pi} \\ &= -(\cos(\pi) - \cos(0)) = -((-1)-1) = 2\,.\end{aligned}$$

74.
$$\int_1^2 \frac{3}{t}dt = 3\times\int_1^2 \frac{1}{t}dt$$
$$\begin{aligned}3\times\int_1^2 \frac{1}{t}dt &= 3\times\ln(t)\Big|_1^2 \\ &= 3\times\ln(2) - 3\times\ln(1) \\ &= 3\times\ln(2) - 3\times 0 \\ &= 3\ln(2)\,.\end{aligned}$$

75.
$$\begin{aligned}\int_2^3 x^{-2} dx &= \frac{1}{(-2+1)}x^{-2+1}\Big|_2^3 = \frac{1}{(-1)}x^{-1}\Big|_2^3 \\ &= (-1)\times\frac{1}{x}\Big|_2^3 = -\frac{1}{x}\Big|_2^3 = -\frac{1}{3} - (-\frac{1}{2}) \\ &= -\frac{1}{3} + \frac{1}{2} = -\frac{2}{6} + \frac{3}{6} \\ &= \frac{-2+3}{6} = \frac{1}{6}\,.\end{aligned}$$

Trigonometric Integrals and Substitutions

Part I: Trigonometric Integrals. We will distinguish three main cases:

Case A: Integrals of type

$$\boxed{\int \sin^m x \cos^n x dx}$$

where m and n are nonnegative integers.

METHOD OF INTEGRATION:
(i) If m is odd, then $u = \cos x$.

(ii) If n is odd, then $u = \sin x$.

(iii) If both m and n are even, then use the identities

$$\boxed{\sin^2 x = \frac{1}{2}(1 - \cos 2x), \quad \cos^2 x = \frac{1}{2}(1 + \cos 2x)}$$

or(and) sometimes

$$\boxed{\sin x \cos x = \frac{1}{2}\sin 2x}$$

76. Find $\int \sin^3 x dx$.

$$\int \sin^3 x dx = \int \sin^2 x \cdot \sin x dx = \int (1 - \cos^2 x) \sin x dx = \begin{bmatrix} \cos x = u \\ d\cos x = du \\ -\sin x dx = du \\ \sin x dx = -du \end{bmatrix}$$

$$= -\int (1 - u^2) du = -u + \frac{1}{3}u^3 + C = -\cos x + \frac{1}{3}\cos^3 x + C$$

77. Find $\int \sin^5 x dx$.

$$\int \sin^5 x dx = \int \sin^4 x \cdot \sin x dx = \int (\sin^2 x)^2 \sin x dx = \int (1 - \cos^2 x)^2 \sin x dx$$

$$= \begin{bmatrix} \cos x = u \\ d\cos x = du \\ -\sin x dx = du \\ \sin x dx = -du \end{bmatrix} = -\int (1 - u^2)^2 du = -\int (1 - 2u^2 + u^4) du$$

$$= -u + \frac{2}{3}u^3 - \frac{1}{5}u^5 + C = -\cos x + \frac{2}{3}\cos^3 x - \frac{1}{5}\cos^5 x + C$$

78. Find $\int \sin^2 x \cos^3 x dx$.

$$\int \sin^2 x \cos^3 x dx = \int \sin^2 x \cos^2 x \cdot \cos x dx = \int \sin^2 x (1 - \sin^2 x) \cos x dx$$

$$= \begin{bmatrix} \sin x = u \\ d\sin x = du \\ \cos x dx = du \end{bmatrix} = \int u^2 (1 - u^2) du = \int (u^2 - u^4) du = \frac{1}{3}u^3 - \frac{1}{5}u^5 + C = \frac{1}{3}\sin^3 x - \frac{1}{5}\sin^5 x + C$$

79. Find $\int \sqrt[3]{\sin x}\cos^5 x dx$.

$$\int \sqrt[3]{\sin x}\cos^5 x dx = \int \sqrt[3]{\sin x}\cos^4 x \cdot \cos x dx = \int \sqrt[3]{\sin x}(\cos^2 x)^2 \cdot \cos x dx$$

$$= \int \sqrt[3]{\sin x}(1-\sin^2 x)^2 \cdot \cos x dx = \begin{bmatrix} \sin x = u \\ d\sin x = du \\ \cos x dx = du \end{bmatrix} = \int \sqrt[3]{u}(1-u^2)^2 du$$

$$= \int u^{1/3}(1-2u^2+u^4)du = \int (u^{1/3}-2u^{2+1/3}+u^{4+1/3})du$$

$$= \int (u^{1/3}-2u^{7/3}+u^{13/3})du = \frac{u^{1/3+1}}{1/3+1} - 2\frac{u^{7/3+1}}{7/3+1} + \frac{u^{13/3+1}}{13/3+1} + C$$

$$= \frac{u^{4/3}}{4/3} - 2\frac{u^{10/3}}{10/3} + \frac{u^{16/3}}{16/3} + C$$

$$= \frac{3}{4}u^{4/3} - \frac{3}{5}u^{10/3} + \frac{3}{16}u^{16/3} + C$$

$$= \frac{3}{4}\sin^{4/3} x - \frac{3}{5}\sin^{10/3} x + \frac{3}{16}\sin^{16/3} x + C$$

80. Find $\int \sin^2 x\cos^2 x dx$.

$$\int \sin^2 x\cos^2 x dx = \int (\sin x\cos x)^2 dx = \int \left(\frac{1}{2}\sin 2x\right)^2 dx = \frac{1}{4}\int \sin^2 2x dx = \frac{1}{4}\int \frac{1}{2}(1-\cos 4x)dx$$

$$= \frac{1}{8}\int (1-\cos 4x)dx = \begin{bmatrix} 4x = u \\ d(4x) = du \\ 4dx = du \\ dx = \frac{1}{4}du \end{bmatrix} = \frac{1}{32}\int (1-\cos u)du = \frac{1}{32}(u-\sin u)+C = \frac{1}{32}(4x-\sin 4x)+C$$

81.

$$\begin{aligned}
\int \sin^5 x\cos^4 x\,dx &= \int \sin^4 x\cos^4 x\sin x\,dx \\
&= \int \left(\sin^2 x\right)^2 \cos^4 x\sin x\,dx \\
&= \int \left(1-\cos^2 x\right)^2 \cos^4 x\sin x\,dx \\
&= -\int \left(1-2\cos^2 x+\cos^4 x\right)\cos^4 x\left(-\sin x\right)dx \\
&= -\int \left[\cos^4 x-2\cos^6 x+\cos^8 x\right]\left(-\sin x\right)dx \\
&= -\left[\frac{1}{5}\cos^5 x-2\frac{1}{7}\cos^7 x+\frac{1}{9}\cos^9 x\right]+c
\end{aligned}$$

82.
$$\begin{aligned}\int \sin^{14} x \cos^3 x dx &= \int \sin^{14} x \cos^2 x \cos x dx \\ &= \int \sin^{14} x \left(1 - \sin^2 x\right) \cos x dx \\ &= \int \left(\sin^{14} x - \sin^{16} x\right) \cos x dx \\ &= \int \sin^{14} x \cos x dx - \int \sin^{16} x \cos x dx\end{aligned}$$

$$\int \sin^{14} x \cos^3 x dx = \frac{\sin^{15} x}{15} - \frac{\sin^{17} x}{17} + c$$

Part II: Trigonometric Substitutions. Here we deal with integrals that involve

$$\boxed{\sqrt{a^2 - x^2}, \quad \sqrt{a^2 + x^2}, \quad \sqrt{x^2 - a^2}}$$

METHOD OF INTEGRATION:

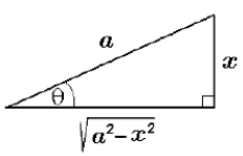
(i) If $\sqrt{a^2 - x^2}$, then $x = a \sin \theta$, $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$.

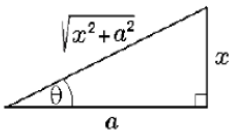
(ii) If $\sqrt{a^2 + x^2}$, then $x = a \tan \theta$, $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$.

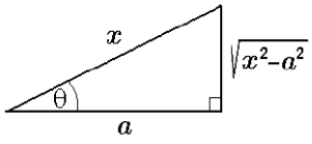
(iii) If $\sqrt{x^2 - a^2}$, then $x = a \sec \theta$, $0 \leq \theta < \frac{\pi}{2}$ or $\pi \leq \theta < \frac{3\pi}{2}$.

Trigonometric Substitutions

$\sqrt{a^2 - x^2}$	$\sqrt{a^2 + x^2}$	$\sqrt{x^2 - a^2}$
$x = a \sin \theta$	$x = a \tan \theta$	$x = a \sec \theta$
$dx = a \cos \theta d\theta$	$dx = a \sec^2 \theta d\theta$	$dx = a \sec \theta \tan \theta d\theta$
$\sqrt{a^2 - x^2} = a \cos \theta$	$\sqrt{a^2 + x^2} = a \sec \theta$	$\sqrt{x^2 - a^2} = a \tan \theta$







83. Find $\displaystyle\int \frac{\sqrt{4-x^2}}{x^2}dx.$

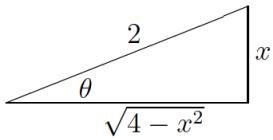
$$\int \frac{\sqrt{4-x^2}}{x^2}dx = \left[\begin{array}{c} x = 2\sin\theta \\ dx = d(2\sin\theta) \\ dx = 2\cos\theta d\theta \\ \sqrt{4-x^2} = \sqrt{4-4\sin^2\theta} = \sqrt{4(1-\sin^2\theta)} = \sqrt{4\cos^2\theta} = 2|\cos\theta| = 2\cos\theta \end{array}\right]$$

$$= \int \frac{2\cos\theta}{4\sin^2\theta}\cdot 2\cos\theta d\theta = \int \frac{\cos^2\theta}{\sin^2\theta}d\theta = \int \frac{1-\sin^2\theta}{\sin^2\theta}d\theta = \int \left(\frac{1}{\sin^2\theta} - \frac{\sin^2\theta}{\sin^2\theta}\right)d\theta$$

$$= \int (\csc^2\theta - 1)d\theta = -\cot\theta - \theta + C$$

Note that

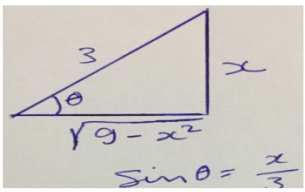
$$\cot\theta = \frac{\sqrt{4-x^2}}{x}$$



Therefore

$$\int \frac{\sqrt{4-x^2}}{x^2}dx = -\frac{\sqrt{4-x^2}}{x} - \sin^{-1}\left(\frac{x}{2}\right) + C$$

84. *Evaluate* $\displaystyle\int \frac{1}{x\sqrt{9-x^2}}dx$



$$dx = 3\cos\theta d\theta$$

$$\sqrt{9-x^2} = \sqrt{9-9\sin^2\theta} = 3\sqrt{1-\sin^2\theta} = 3\cos\theta$$

$$\begin{aligned} \int \frac{1}{x\sqrt{9-x^2}}dx &= \int \frac{1}{(3\sin\theta)(3\cos\theta)}(3\cos\theta)\,d\theta \\ &= \int \frac{1}{3\sin\theta}d\theta \\ &= \frac{1}{3}\ln|\csc\theta - \cot\theta| + c \end{aligned}$$

$$\int \frac{1}{x\sqrt{9-x^2}}dx = \frac{1}{3}\ln\left|\frac{3}{x} - \frac{\sqrt{9-x^2}}{x}\right| + c$$

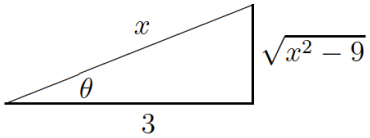
85. Find $\displaystyle\int \frac{1}{x^2\sqrt{x^2-9}}dx.$

$$\int \frac{1}{x^2\sqrt{x^2-9}}dx = \left[\begin{array}{c} x = 3\sec\theta \\ dx = d(3\sec\theta) \\ dx = 3\sec\theta\tan\theta d\theta \\ \sqrt{x^2-9} = \sqrt{9\sec^2\theta - 9} = \sqrt{9(\sec^2\theta - 1)} = \sqrt{9\tan^2\theta} = 3|\tan\theta| = 3\tan\theta \end{array}\right]$$

$$= \int \frac{1}{9\sec^2\theta\cdot 3\tan\theta}\cdot 3\sec\theta\tan\theta d\theta = \frac{1}{9}\int \cos\theta d\theta = \frac{1}{9}\sin\theta + C$$

Note that

$$\sin\theta = \frac{\sqrt{x^2-9}}{x}$$



Therefore

$$\int \frac{1}{x^2\sqrt{x^2-9}}dx = \frac{1}{9}\cdot\frac{\sqrt{x^2-9}}{x} + C$$

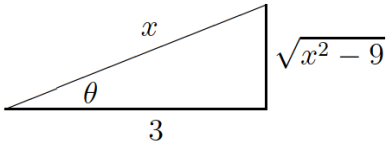
86. Find $\int \frac{x}{\sqrt{x^2-9}}dx$.

$$\int \frac{x}{\sqrt{x^2-9}}dx = \left[\begin{array}{c} x = 3\sec\theta \\ dx = d(3\sec\theta) \\ dx = 3\sec\theta\tan\theta d\theta \\ \sqrt{x^2-9} = \sqrt{9\sec^2\theta - 9} = \sqrt{9(\sec^2\theta - 1)} = \sqrt{9\tan^2\theta} = 3|\tan\theta| = 3\tan\theta \end{array}\right]$$

$$= \int \frac{3\sec\theta}{3\tan\theta}\cdot 3\sec\theta\tan\theta d\theta = 3\int \sec^2\theta d\theta = 3\tan\theta + C$$

Note that

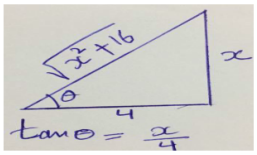
$$\tan\theta = \frac{\sqrt{x^2-9}}{3}$$



Therefore

$$\int \frac{x}{\sqrt{x^2-9}}dx = 3\frac{\sqrt{x^2-9}}{3} = \sqrt{x^2-9} + C$$

87. $\int \frac{x^2}{(x^2+16)^{\frac{3}{2}}}dx$



$x = 4\tan\theta \qquad dx = 4\sec^2\theta d\theta$

$$\begin{aligned}\int \frac{x^2}{(x^2+16)^{\frac{3}{2}}}dx &= \int \frac{(16\tan^2\theta)}{(64\sec^3\theta)}(4\sec^2\theta)\,d\theta \\ &= \int \frac{\tan^2\theta}{\sec\theta}d\theta \\ &= \int \frac{\sec^2\theta - 1}{\sec\theta}d\theta \\ &= \int \sec\theta - \cos\theta d\theta \\ &= \ln|\sec\theta + \tan\theta| - \sin\theta + c\end{aligned}$$

$$\int \frac{x^2}{(x^2+16)^{\frac{3}{2}}}dx == \ln\left|\frac{\sqrt{x^2+16}}{4} + \frac{x}{4}\right| - \frac{x}{\sqrt{x^2+16}} + c$$

88. Evaluate $\int \sqrt{16-x^2}dx$.

Let $x = 4\sin\theta$, $dx = 4\cos\theta d\theta$, $\sqrt{16-x^2} = 4\cos\theta$. Then

$$\begin{aligned}\int \sqrt{16-x^2}dx &= \int (4\cos\theta)(4\cos\theta)d\theta \\ &= 16\int \cos^2\theta d\theta \\ &= 8\int (1+\cos 2\theta)d\theta \\ &= 8\left(\theta + \frac{1}{2}\sin 2\theta\right) + C \\ &= 8\theta + 8\sin\theta\cos\theta + C \\ &= 8\sin^{-1}\left(\frac{x}{4}\right) + 8\left(\frac{x}{4}\right)\left(\frac{\sqrt{16-x^2}}{4}\right) + C \\ &= 8\sin^{-1}\left(\frac{x}{4}\right) + \frac{x\sqrt{16-x^2}}{2} + C.\end{aligned}$$

89. $$\int \frac{x^2}{\left(4-x^2\right)^{3/2}}\,dx = \int \frac{x^2}{\left(\sqrt{4-x^2}\right)^3}\,dx$$

Use the trigonometric substitution $x = 2\sin\theta$ to get $dx = 2\cos\theta\,d\theta$. These substitutions give you the following:

$$\begin{aligned}\int \frac{(2\sin\theta)^2 2\cos\theta\,d\theta}{\left(\sqrt{4-4\sin^2\theta}\right)^3} &= \int \frac{8\sin^2\theta\cos\theta\,d\theta}{\left(2\sqrt{\cos^2\theta}\right)^3}\\ &= \int \frac{\sin^2\theta\cos\theta}{\cos^3\theta}\,d\theta\\ &= \int \tan^2\theta\,d\theta\\ &= \int\left(\sec^2\theta - 1\right)d\theta\\ &= \tan\theta - \theta + C\end{aligned}$$

From the substitution $x = 2\sin\theta$, or $\frac{x}{2} = \sin\theta$, you can deduce that $\sin^{-1}\left(\frac{x}{2}\right) = \theta$ and that $\tan\theta = \frac{x}{\sqrt{4-x^2}}$. (To deduce that $\tan\theta = \frac{x}{\sqrt{4-x^2}}$, you can label a right triangle using $\frac{x}{2} = \sin\theta$ and then find the missing side with the Pythagorean theorem.) Therefore, the solution becomes

$$\tan\theta - \theta + C = \frac{x}{\sqrt{4-x^2}} - \sin^{-1}\left(\frac{x}{2}\right) + C$$

90. $$\int \frac{\sqrt{x^2-1}}{x}\,dx$$

Begin with the substitution $= \sec\theta$ so that $dx = \sec\theta\tan\theta\,d\theta$:

$$\begin{aligned}&\int \frac{\sqrt{\sec^2\theta - 1}}{\sec\theta}\sec\theta\tan\theta\,d\theta\\ &= \int \sqrt{\tan^2\theta}\,\tan\theta\,d\theta\\ &= \int \tan^2\theta\,d\theta\\ &= \int\left(\sec^2\theta - 1\right)d\theta = \tan\theta - \theta + C\end{aligned}$$

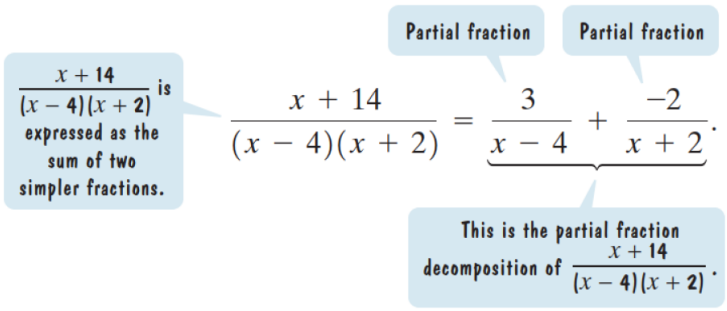
From the substitution $x = \sec\theta$, you can deduce that $\sec^{-1}x = \theta$ and that $\tan\theta = \sqrt{x^2-1}$ to get the solution:

$$= \sqrt{x^2-1} - \sec^{-1}x + C$$

The Idea behind Partial Fraction Decomposition

We know how to use common denominators to write a sum or difference of rational expressions as a single rational expression. For example,

$$\begin{aligned}\frac{3}{x-4} - \frac{2}{x+2} &= \frac{3(x+2) - 2(x-4)}{(x-4)(x+2)}\\ &= \frac{3x + 6 - 2x + 8}{(x-4)(x+2)} = \frac{x+14}{(x-4)(x+2)}.\end{aligned}$$



The Partial Fraction Decomposition of a Rational Expression with Distinct Linear Factors in the Denominator

Case 1: Distinct Linear Factors

$$\frac{R(x)}{Q(x)} = \frac{R(x)}{(x-a_1)\cdots(x-a_n)} = \frac{A_1}{x-a_1} + \cdots + \frac{A_n}{x-a_n}$$

91.

$$\frac{x+8}{x^2+x-2} = \frac{x+8}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$$

Summing the right hand side, we obtain

$$\frac{x+8}{(x-1)(x+2)} = \frac{A(x+2)+B(x-1)}{(x-1)(x+2)}$$

$$x+8 = A(x+2)+B(x-1)$$

$$x=1: \quad 9=3A \implies A=3$$
$$x=-2: \quad 6=-3B \implies B=-2$$

$$\int \frac{x+8}{x^2+x-2}\,dx = \int \frac{3}{x-1} - \frac{2}{x+2}\,dx = 3\ln|x-1| - 2\ln|x+2| + c$$

92.

$$\int \frac{x+5}{x^2-2x-8}\,dx$$

$$\frac{x+5}{x^2-2x-8} = \frac{x+5}{(x-4)(x+2)} = \frac{A}{x-4} + \frac{B}{x+2}$$

$$A = \lim_{x\to 4} \frac{x+5}{x+2} = \frac{9}{6} = \frac{3}{2} \qquad B = \lim_{x\to -2} \frac{x+5}{x-4} = \frac{3}{-6} = \frac{-1}{2}$$

$$\int \frac{x+5}{x^2-2x-8}\,dx = \int \left(\frac{3/2}{x-4} + \frac{-1/2}{x+2}\right) dx$$

$$= \frac{3}{2}\ln|x-4| - \frac{1}{2}\ln|x+2| + C$$

93.
$$\int \frac{x-2}{x(x-3)(x+1)}\,dx$$

$$\frac{x-2}{x(x-3)(x+1)} = \frac{A}{x} + \frac{B}{x-3} + \frac{C}{x+1}$$

$$A = \lim_{x\to 0} \frac{x-2}{(x-3)(x+1)} = \frac{2}{3}$$

$$B = \lim_{x\to 3} \frac{x-2}{x(x+1)} = \frac{1}{12}$$

$$C = \lim_{x\to -1} \frac{x-2}{x(x-3)} = \frac{-3}{4}$$

$$\int \frac{x-2}{x(x-3)(x+1)}\,dx = \int \left(\frac{2/3}{x} + \frac{1/12}{x-3} + \frac{-3/4}{x+1} \right) dx$$

$$= \frac{2}{3}\ln|x| + \frac{1}{12}\ln|x-3| - \frac{3}{4}\ln|x+1| + C$$

94.
$$\int \frac{x-3}{(x+4)(x-5)}\,dx$$

First perform a fraction decomposition on the integrand:

$$\frac{x-3}{(x+4)(x-5)} = \frac{A}{x+4} + \frac{B}{x-5}$$

Multiplying both sides of the equation by $(x + 4)(x – 5)$ and simplifying gives you

$$(x+4)(x-5)\left(\frac{x-3}{(x+4)(x-5)} \right) = (x+4)(x-5)\left(\frac{A}{x+4} + \frac{B}{x-5} \right)$$

$$x-3 = A(x-5) + B(x+4)$$

Now you can find the values of the coefficients by picking appropriate values of x and solving the resulting equations. So if you let $x = 5$, you get $5 – 3 = A(5 – 5) + B(5 + 4)$, or $2 = 9B$, so that $\frac{2}{9} = B$. Likewise, if $x = –4$, then you get $–4 – 3 = A(–4 – 5) + B(–4 + 4)$, or $–7 = A(–9)$, so that $\frac{7}{9} = A$. This gives you

$$\int \frac{x-3}{(x+4)(x-5)}\,dx = \int \left(\frac{A}{x+4} + \frac{B}{x-5} \right) dx$$

$$= \int \left(\frac{7/9}{x+4} + \frac{2/9}{x-5} \right) dx$$

Applying elementary antiderivative formulas gives you the solution:

$$\frac{7}{9}\ln|x+4| + \frac{2}{9}\ln|x-5| + C$$

Improper Integrals

Improper integrals with an infinite limit of integration:

1. **If $f(x)$ is continuous on $[a, \infty)$, then**

$$\int_a^{\infty} f(x)\,dx = \lim_{t\to\infty} \int_a^t f(x)\,dx$$

provided this limit exists as a finite number.

2. **If $f(x)$ is continuous on $(-\infty, b]$, then**

$$\int_{-\infty}^{b} f(x)\,dx = \lim_{t\to-\infty} \int_t^b f(x)\,dx$$

provided this limit exists as a finite number.

Remark 2.1 *The improper integrals in (1) and (2) are called convergent if the limit exists and divergent if the limit does not exist.*

3. If both $\int_a^{\infty} f(x)\,dx$ and $\int_{-\infty}^{a} f(x)\,dx$ are convergent, then define

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{a} f(x)\,dx + \int_a^{\infty} f(x)\,dx$$

95. *Determine whether the integral* $\int_4^{\infty} \frac{1}{(x-1)^3}dx$ *converges or diverges and if it is converges, find its value.*

$$\begin{aligned}
\int_4^{\infty} \frac{1}{(x-1)^3}dx &= \lim_{t\to\infty} \int_4^t \frac{1}{(x-1)^3}dx \\
&= \lim_{t\to\infty} \left[\frac{-1}{2(x-1)^2}\right]_4^t \\
&= \lim_{t\to\infty} \left[\frac{-1}{2(t-1)^2} + \frac{1}{2(4-1)^2}\right] \\
&= 0 + \frac{1}{18} = \frac{1}{18}
\end{aligned}$$

The integral is convergent. .

96. *Determine whether the integral* $\int_4^{\infty} \frac{1}{x-1}dx$ *converges or diverges and if it is converges, find its value.*

$$\begin{aligned}\int_4^\infty \frac{1}{x-1}dx &= \lim_{t\to\infty}\int_4^t \frac{1}{x-1}dx \\ &= \lim_{t\to\infty}\left[\ln(x-1)\right]_4^t \\ &= \lim_{t\to\infty}\left[\ln(t-1)-\ln(4-1)\right] \\ &= \lim_{t\to\infty}\ln(t-1)-\ln(3) \\ &= \infty\end{aligned}$$

It is clear that the limit does not exist and hence the integral diverges.

97. $$\int_{-\infty}^{0} e^{3x}\,dx$$

$$\begin{aligned}\int_{-\infty}^0 e^{3x}\,dx &= \lim_{t\to-\infty}\left[\int_t^0 e^{3x}\,dx\right] \\ &= \lim_{t\to-\infty}\left[\frac{1}{3}e^{3x}\right]_t^0 \\ &= \lim_{t\to-\infty}\left(\frac{1}{3}-\frac{1}{3}e^{3t}\right)=\frac{1}{3}\end{aligned}$$

$\int_{-\infty}^0 e^{3x}\,dx$ is convergent.

98. Find

$$\int_0^\infty e^{-x}\,dx.$$

(if it even converges)

Solution:

$$\begin{aligned}\int_0^\infty e^{-x}\,dx &= \lim_{b\to\infty}\int_0^b e^{-x}\,dx = \lim_{b\to\infty}\left[-e^{-x}\right]_0^b \\ &= \lim_{b\to\infty} -e^{-b}+e^0 = 0+1=1.\end{aligned}$$

So the integral converges and equals 1.

99. $$\int_1^\infty \frac{1}{(x+1)^2}dx = \lim_{a\to\infty}\int_1^a \frac{1}{(x+1)^2}dx$$

First evaluate $\int \frac{1}{(x+1)^2}dx$ using the substitution $u = x + 1$ so that $du = dx$. Using these substitutions gives you

$$\begin{aligned}\int \frac{1}{u^2}du &= \int u^{-2}du \\ &= -u^{-1}+C \\ &= -\frac{1}{x+1}+C\end{aligned}$$

$$\lim_{a\to\infty}\int_1^a \frac{1}{(x+1)^2}\,dx = \lim_{a\to\infty}\left(-\frac{1}{x+1}\Big|_1^a\right)$$

Now evaluate the limit:

$$\begin{aligned}
&\lim_{a\to\infty}\left(-\frac{1}{x+1}\Big|_1^a\right)\\
&=\lim_{a\to\infty}\left(-\frac{1}{a+1}+\frac{1}{1+1}\right)\\
&=\left(0+\frac{1}{2}\right)\\
&=\frac{1}{2}
\end{aligned}$$

Because the value of the integral is finite, the improper integral is convergent.

100. $$\int_{-\infty}^1 e^{-4x}dx = \lim_{a\to-\infty}\int_a^1 e^{-4x}dx$$

Then use elementary antiderivatives to get the follow

$$\begin{aligned}
\int_{-\infty}^1 e^{-4x}dx &= \lim_{a\to-\infty}\int_a^1 e^{-4x}dx\\
&= \lim_{a\to-\infty}\left(-\frac{1}{4}e^{-4x}\Big|_a^1\right)\\
&= \lim_{a\to-\infty}\left(-\frac{1}{4}e^{-4(1)}-\left(-\frac{1}{4}e^{-4a}\right)\right)
\end{aligned}$$

Because $\lim_{a\to-\infty} e^{-4a}=\infty$, the integral is divergent.

101. $$\int_2^\infty e^{-x/3}dx = \lim_{a\to\infty}\int_2^a e^{-x/3}dx$$

To evaluate $\int e^{-x/3}dx$, let $u=-\frac{1}{3}x$ so that $du=-\frac{1}{3}dx$, or $-3\,du = dx$. This gives you

$$\begin{aligned}
\int e^{-x/3}dx &= -3\int e^u du\\
&= -3e^u + C\\
&= -3e^{-x/3} + C
\end{aligned}$$

Because $\int e^{-x/3}dx = -3e^{-x/3}+C$, it follows that

$$\lim_{a\to\infty}\int_2^a e^{-x/3}dx = \lim_{a\to\infty}\left(-3e^{-x/3}\Big|_2^a\right)$$

Now simply evaluate the limit:

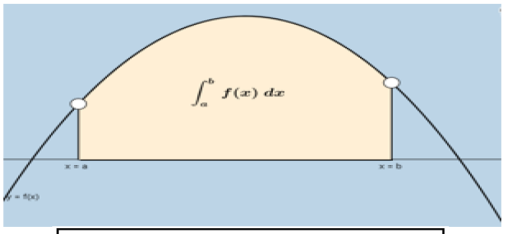
$$\begin{aligned}
\lim_{a\to\infty}\left(-3e^{-x/3}\Big|_2^a\right) &= \lim_{a\to\infty}\left(-3e^{-a/3}+3e^{-2/3}\right)\\
&= -3e^{-\infty/3}+3e^{-2/3}\\
&= 0+3e^{-2/3}\\
&= 3e^{-2/3}
\end{aligned}$$

The answer is finite, so the integral is convergent.

Application of Definite Integrals

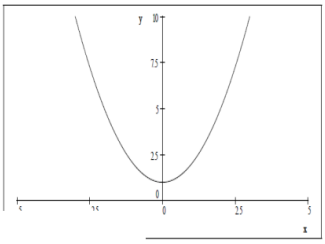
1) Area:

a) **Area Under a Curve**



$$A = \int_a^b f\ (x)\ dx$$

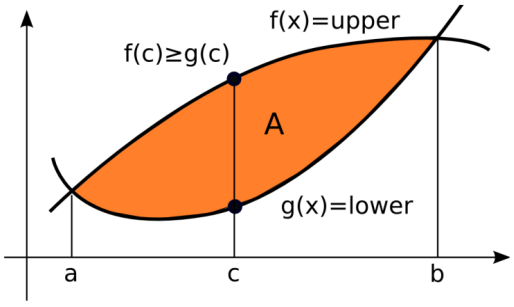
102. Example: Find the area under the curve $y = x^2 + 1$ from $x = 1$ and $x = 3$.



Solution. The area is given by

$$\begin{aligned} A &= \int_1^3 x^2 + 1 dx \\ &= \left[\frac{x^3}{3} + x\right]_1^3 \\ &= \frac{32}{3} \end{aligned}$$

Areas between Curves

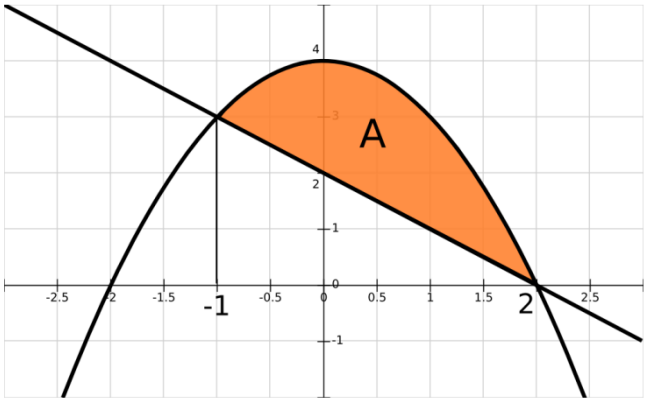


Area between two curves $= \int_a^b$ (upper curve - lower curve) dx

If $f(x) \geq g(x)$ (as on the figure above), then the area is $A = \int_a^b (f(x) - g(x))\ dx.$

103. Find the area enclosed by the curves $f(x) = 4 - x^2$ and $g(x) = 2 - x$.

Solution. Graph both curves first and note that they intersect two times. These intersections are the bounds of the integration.



Find the intersections by solving

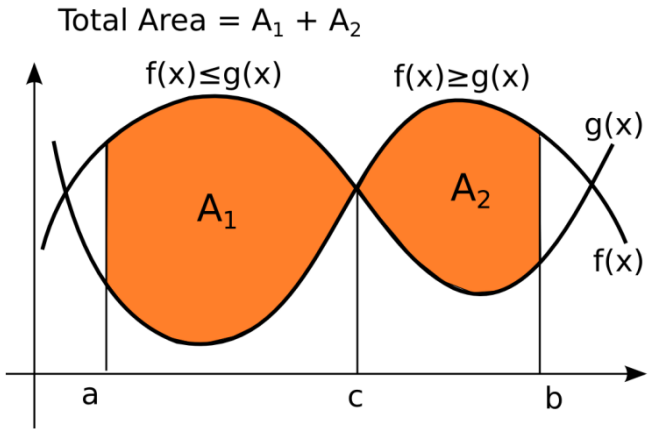
$$4 - x^2 = 2 - x \Rightarrow x^2 - x - 2 = 0 \Rightarrow$$

$$(x - 2)(x + 1) = 0 \Rightarrow x = 2 \text{ and } x = 1.$$

The graph indicates that the curve $4 - x^2$ is upper and $2 - x$ is lower. You can double check that by plugging a number between -1 and 2 into both. For example, using $x = 0$ we have that $f(0) = 4 > g(0) = 2$.

$$A = \int_{-1}^{2} \left(4 - x^2 - (2 - x)\right)\, dx = \int_{-1}^{2} \left(2 - x^2 + x\right)\, dx = \left(2x - \frac{x^3}{3} + \frac{x^2}{2}\right)\Big|_{-1}^{2} =$$

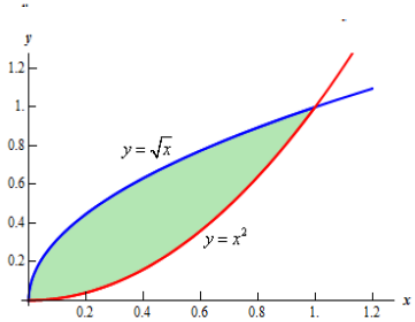
$$\left(2(2) - \frac{2^3}{3} + \frac{2^2}{2}\right) - \left(2(-1) - \frac{(-1)^3}{3} + \frac{(-1)^2}{2}\right) = 5 - \frac{1}{2} = \frac{9}{2} = 4.5.$$



$$A = A_1 + A_2 = \int_a^c (g(x) - f(x))\, dx + \int_c^b (f(x) - g(x))\, dx.$$

104. Determine the area of the region enclosed by $y = x^2$ and $y = \sqrt{x}$.

$$A = \int_a^b \left(\begin{matrix}\text{upper}\\ \text{function}\end{matrix}\right) - \left(\begin{matrix}\text{lower}\\ \text{function}\end{matrix}\right) dx$$
$$= \int_0^1 \sqrt{x} - x^2\, dx$$
$$= \left(\frac{2}{3}x^{\frac{3}{2}} - \frac{1}{3}x^3\right)\Big|_0^1$$
$$= \frac{1}{3}$$

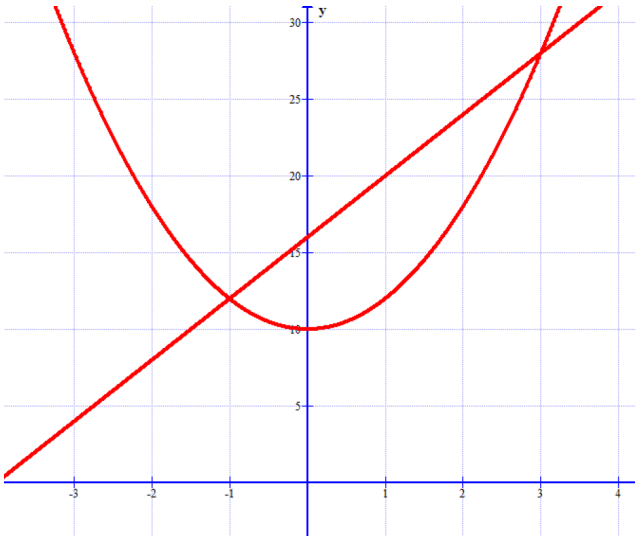


$x^2 = \sqrt{x}$
$x^4 = x$
$x^4 - x = 0$
$x(x^3 - 1) = 0$
$x = 0, \quad x = 1$

105. Determine the area of the region bounded by $y = 2x^2 + 10$ and $y = 4x + 16$.

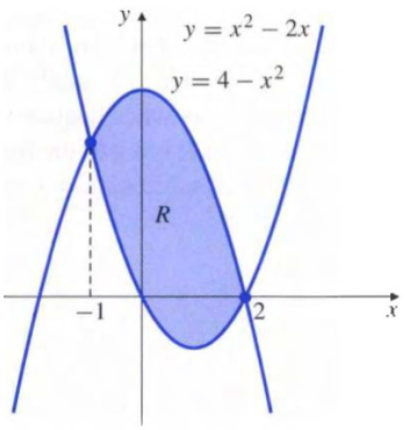
The two curves will intersect at $x = -1$ and $x = 3$.

$$A = \int_a^b \left(\begin{matrix}\text{upper}\\ \text{function}\end{matrix}\right) - \left(\begin{matrix}\text{lower}\\ \text{function}\end{matrix}\right) dx$$
$$= \int_{-1}^3 4x + 16 - (2x^2 + 10)\, dx$$
$$= \int_{-1}^3 -2x^2 + 4x + 6\, dx$$
$$= \left(-\frac{2}{3}x^3 + 2x^2 + 6x\right)\Big|_{-1}^3$$
$$= \frac{64}{3}$$



106. Find the area of the bounded, plane region R lying between the curves $y = x^2 - 2x$ and $y = 4 - x^2$.

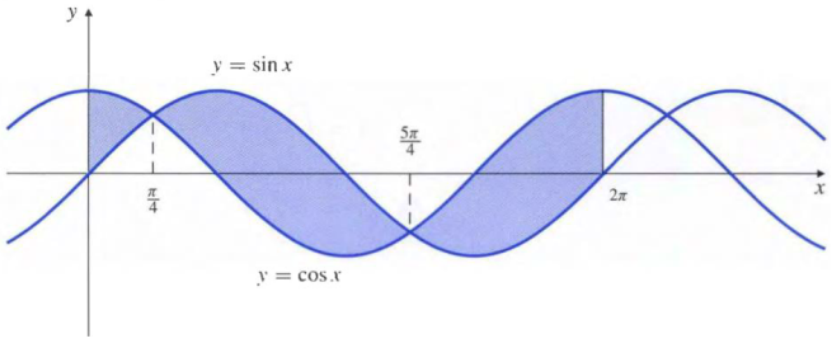
Solution First, we must find the intersections of the curves, so we solve the equations simultaneously:



$$x^2 - 2x = y = 4 - x^2$$
$$2x^2 - 2x - 4 = 0$$
$$2(x-2)(x+1) = 0 \quad \text{so } x = 2 \text{ or } x = -1.$$

$$A = \int_{-1}^{2} \left((4 - x^2) - (x^2 - 2x)\right) dx$$
$$= \int_{-1}^{2} (4 - 2x^2 + 2x)\, dx$$
$$= \left(4x - \frac{2}{3}x^3 + x^2\right)\Big|_{-1}^{2}$$
$$= 4(2) - \frac{2}{3}(8) + 4 - \left(-4 + \frac{2}{3} + 1\right) = 9 \text{ square units.}$$

107. Find the total area A lying between the curves $y = \sin x$ and $y = \cos x$ from $x = 0$ to $x = 2\pi$.



$$A = \int_0^{\pi/4} (\cos x - \sin x)\, dx + \int_{\pi/4}^{5\pi/4} (\sin x - \cos x)\, dx$$
$$+ \int_{5\pi/4}^{2\pi} (\cos x - \sin x)\, dx$$
$$= (\sin x + \cos x)\Big|_0^{\pi/4} - (\cos x + \sin x)\Big|_{\pi/4}^{5\pi/4} + (\sin x + \cos x)\Big|_{5\pi/4}^{2\pi}$$
$$= (\sqrt{2} - 1) + (\sqrt{2} + \sqrt{2}) + (1 + \sqrt{2}) = 4\sqrt{2} \text{ square units.}$$

108. find the area of the plane region bounded by the given curves.

$$y = x^2 - 2x, \quad y = 6x - x^2$$

For intersections:

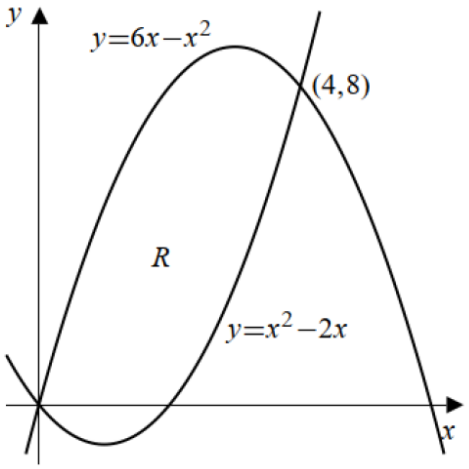
$x^2 - 2x = 6x - x^2 \Rightarrow 2x^2 - 8x = 0$

i.e., $x = 0$ or 4.

$$\text{Area of } R = \int_0^4 \left[6x - x^2 - (x^2 - 2x)\right] dx$$

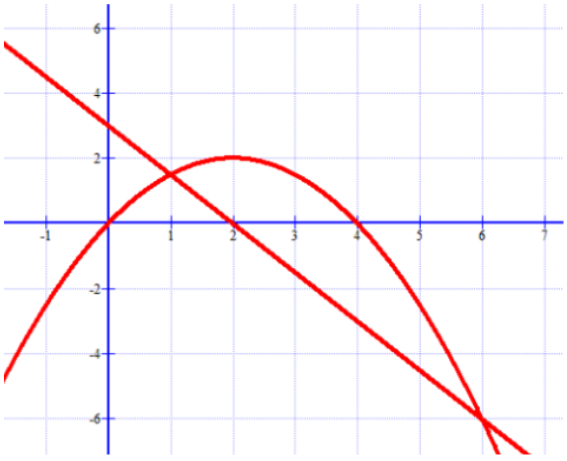
$$= \int_0^4 (8x - 2x^2)\, dx$$

$$= \left(4x^2 - \frac{2}{3}x^3\right)\Big|_0^4 = \frac{64}{3} \text{ sq. units.}$$



109. find the area of the plane region bounded by the given curves.

$$2y = 4x - x^2, \quad 2y + 3x = 6$$



$$\left.\begin{aligned} 2y &= 4x - x^2 \\ 2y + 3x &= 6 \end{aligned}\right\} \Rightarrow \begin{aligned} 4x - x^2 &= 6 - 3x \\ x^2 - 7x + 6 &= 0 \\ (x - 1)(x - 6) &= 0 \end{aligned}$$

Thus intersections of the curves occur at $x = 1$ and $x = 6$. We have

$$\text{Area of } R = \int_1^6 \left(2x - \frac{x^2}{2} - 3 + \frac{3x}{2}\right) dx$$

$$= \left(\frac{7x^2}{4} - \frac{x^3}{6} - 3x\right)\Big|_1^6$$

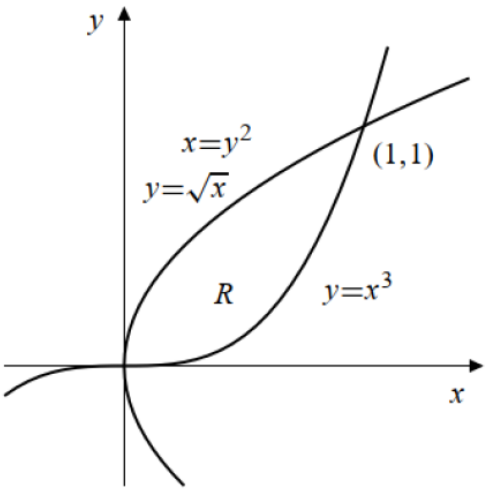
$$= \frac{245}{4} - 36 + \frac{1}{6} - 15 = \frac{125}{12} \text{ sq. units.}$$

110. find the area of the plane region bounded by the given curves.

$$y = x^3, \quad x = y^2$$

$$\text{Area of } R = \int_0^1 (\sqrt{x} - x^3)\, dx$$

$$= \left(\frac{2}{3}x^{3/2} - \frac{x^4}{4}\right)\Big|_0^1 = \frac{5}{12} \text{ sq. units.}$$



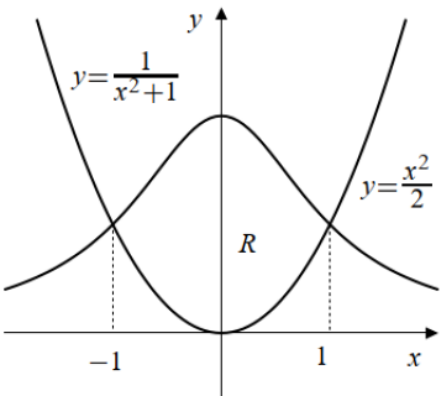
111. find the area of the plane region bounded by the given curves.

$$y = \frac{1}{2}x^2, \quad y = \frac{1}{x^2 + 1}$$

The curves $y = \dfrac{x^2}{2}$ and $y = \dfrac{1}{1+x^2}$ intersect at $x = \pm 1$.
Thus

$$\text{Area of } R = 2\int_0^1 \left(\frac{1}{1+x^2} - \frac{x^2}{2}\right) dx$$

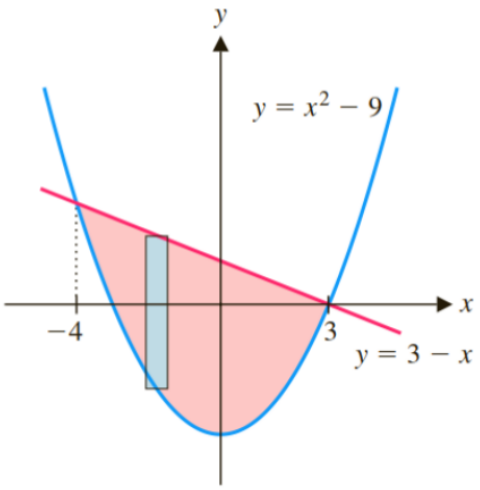
$$= 2\left(\tan^{-1} x - \frac{x^3}{6}\right)\Big|_0^1 = \frac{\pi}{2} - \frac{1}{3} \text{ sq. units.}$$



112. Find the area bounded by the graphs of $y = 3 - x$ and $y = x^2 - 9$

$$3 - x = x^2 - 9 \quad \text{or} \quad 0 = x^2 + x - 12 = (x - 3)(x + 4).$$

Thus, the curves intersect at $x = -4$ and $x = 3$.



$$A = \int_{-4}^{3} [(3 - x) - (x^2 - 9)]\,dx$$

$$= \int_{-4}^{3} (-x^2 - x + 12) = \left[-\frac{x^3}{3} - \frac{x^2}{2} + 12x \right]_{-4}^{3}$$

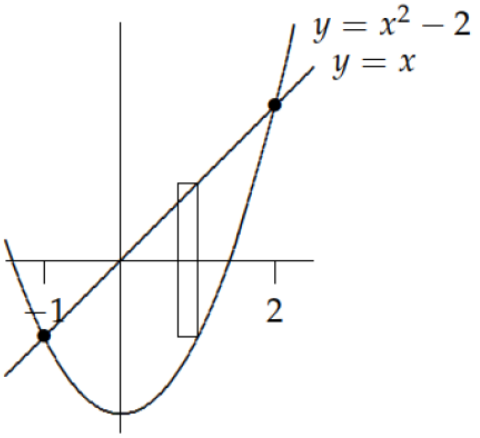
$$= \left[-\frac{3^3}{3} - \frac{3^2}{2} + 12(3) \right] - \left[-\frac{(-4)^3}{3} - \frac{(-4)^2}{2} + 12(-4) \right] = \frac{343}{6}.$$

113. Find the area of the region enclosed by the graphs

$$y = x^2 - 2 \text{ and } y = x.$$

$$x^2 - 2 = x \Rightarrow x^2 - x - 2 = 0 \Rightarrow (x + 1)(x - 2) = 0 \Rightarrow x = -1, 2.$$

Area enclosed by g and $f = \int_a^b [g(x) - f(x)]\,dx = \int_{-1}^{2} [x - (x^2 - 2)]\,dx$

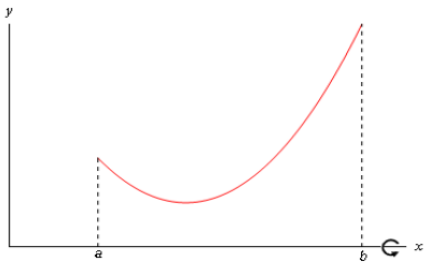


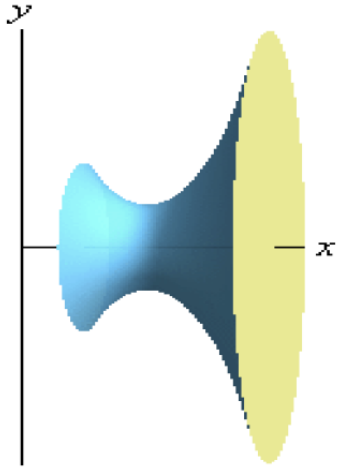
$$= \frac{x^2}{2} - \frac{x^3}{3} + 2x \Big|_{-1}^{2}$$

$$= \left(2 - \frac{8}{3} + 4\right) - \left(\frac{1}{2} + \frac{1}{3} - 2\right) = \frac{9}{2}.$$

2) Volume of Revolution:

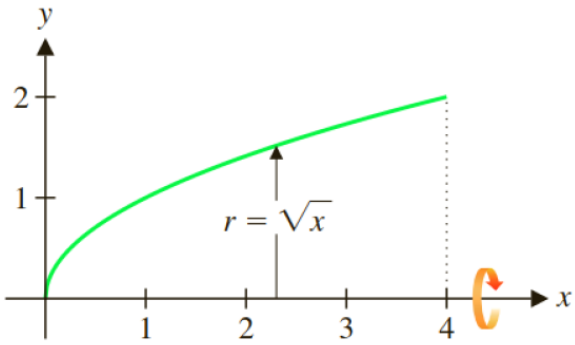
We should first define just what a solid of revolution is. To get a solid of revolution we start out with a function, $y = f(x)$, on an interval $[a, b]$.

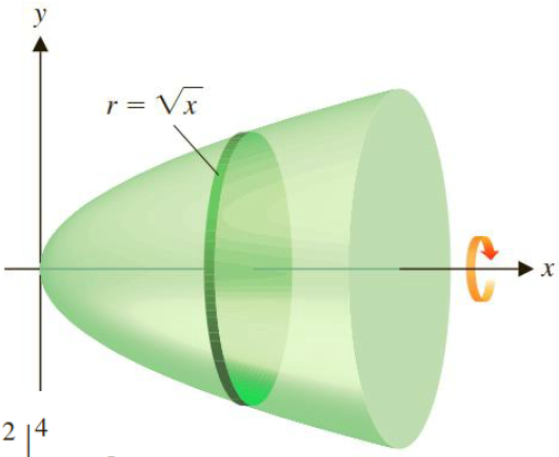




$$V = \pi \int_a^b [f(x)]^2 \, dx$$

114. Revolve the region under the curve $y = \sqrt{x}$ on the interval $[0, 4]$ about the x-axis and find the volume of the resulting solid of revolution.





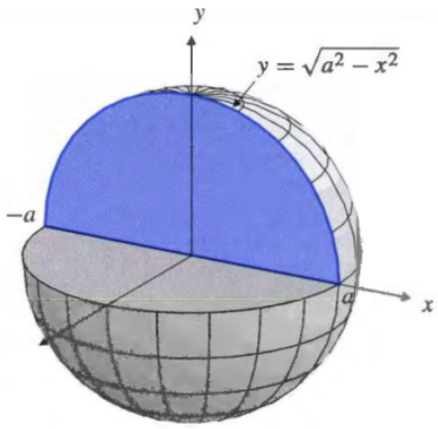
$$V = \int_0^4 \underbrace{\pi\left[\sqrt{x}\right]^2}_{\text{cross-sectional area } = \pi r^2} dx = \pi \int_0^4 x \, dx = \pi \left.\frac{x^2}{2}\right|_0^4 = 8\pi.$$

115. **(The volume of a ball)** Find the volume of a solid ball having radius a.

Solution The ball can be generated by rotating the half-disk, $0 \le y \le \sqrt{a^2 - x^2}$, $-a \le x \le a$ about the x-axis. See the cutaway view in Figure 7.5(a). Therefore, its volume is

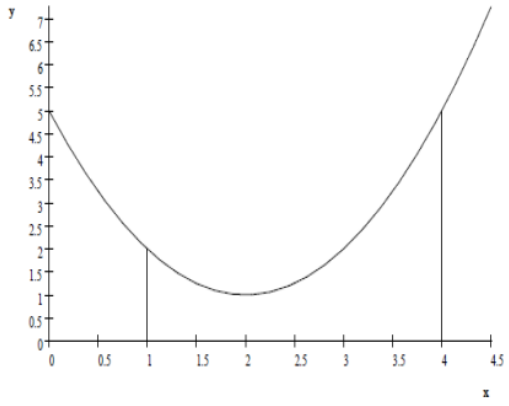
$$V = \pi \int_{-a}^{a} (\sqrt{a^2 - x^2})^2 \, dx = 2\pi \int_{0}^{a} (a^2 - x^2) \, dx$$

$$= 2\pi \left(a^2 x - \frac{x^3}{3} \right) \Big|_0^a = 2\pi \left(a^3 - \frac{1}{3} a^3 \right) = \frac{4}{3} \pi a^3 \text{ cubic uni}$$



116. Determine the volume of the solid obtained by rotating the region bounded by $y = x^2 - 4x + 5$, $x = 1$, $x = 4$, and the x-axis about the x-axis.

Solution:



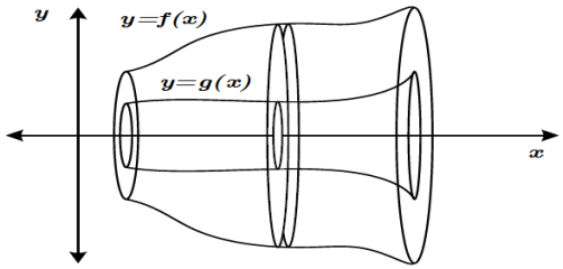
$$V = \pi \int_a^b y^2 dx$$

$$= \pi \int_1^4 \left[x^2 - 4x + 5 \right]^2 dx$$

$$= \pi \int_1^4 \left[x^4 - 8x^3 + 26x^2 - 40x + 25 \right] dx$$

$$= \pi \left[\frac{1}{5} x^5 - 2x^4 + \frac{26}{3} x^3 - 20x^2 + 25x \right]_1^4 = \frac{78\pi}{5}$$

$$V = \pi \int_a^b [f(x)]^2 - [g(x)]^2 \, dx$$



117. The region enclosed by the curves $y = x^2$ and $y = x$ is rotated about the x-axis. Find the volume of the resulting solid.

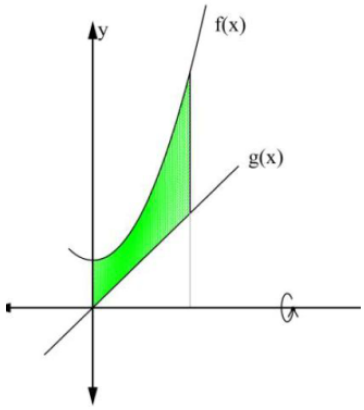
$$x^2 = x \Rightarrow x = 0;\ 1$$

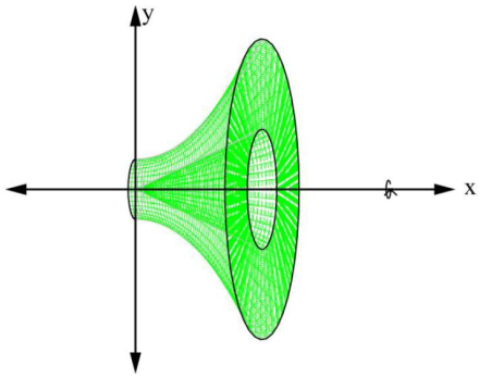
$$\begin{aligned} V &= \pi \int_a^b [f(x)]^2 - [g(x)]^2 \, dx \\ &= \pi \int_0^1 x^2 - \left[x^2\right]^2 dx \\ &= \pi \left[\frac{1}{3}x^3 - \frac{1}{5}x^5\right]_0^1 = \frac{2\pi}{15} \end{aligned}$$

118.

Find the volume generated when the region between the graphs $f(x) = x^2 + 1$ and $g(x) = x$ over the interval $[0, 3]$ is revolved about the $x-$axis.

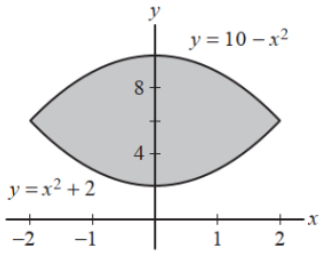
$$\begin{aligned} V(x) &= \int_a^b \pi \left([f(x)]^2 - [g(x)]^2\right) dx \\ &= \int_0^3 \pi \left((x^2+1)^2 - (x)^2\right) dx \\ &= \int_0^3 \pi \left(x^4 + x^2 + 1\right) dx \\ &= \frac{303\pi}{5}. \end{aligned}$$





119. Find the volume generated when the region between the graphs $y = x^2 + 2, \quad y = 10 - x^2$ revolved about the x—axis.

(a) Setting $x^2 + 2 = 10 - x^2$ yields $2x^2 = 8$, or $x^2 = 4$. The two curves therefore intersect at $x = \pm 2$. The region enclosed by the two curves is shown in the figure below.

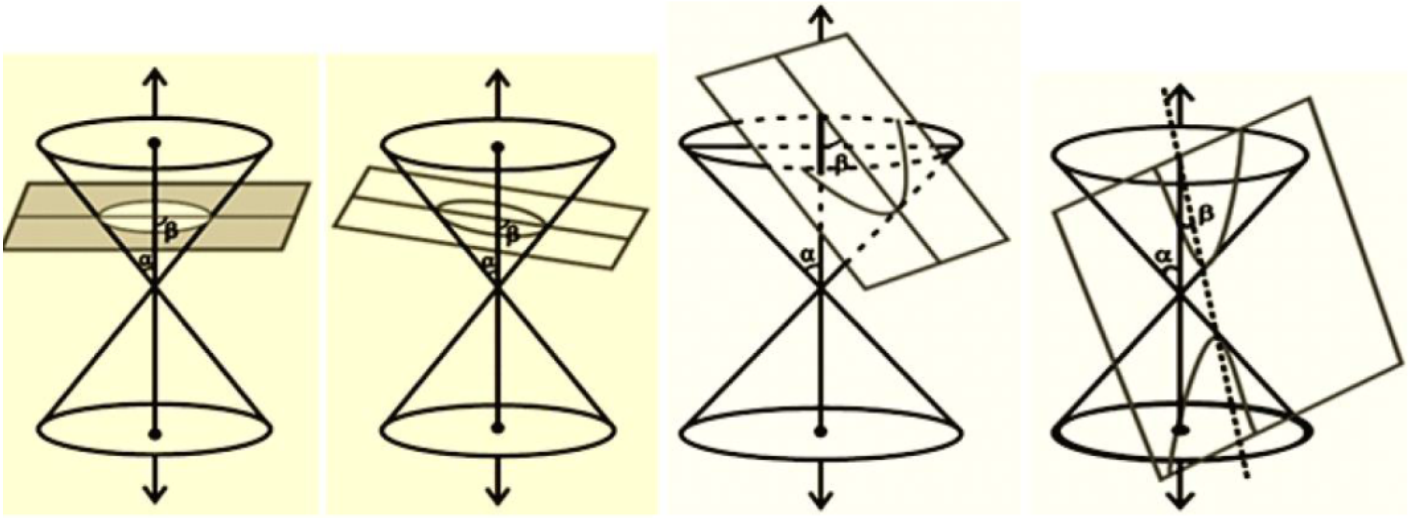


(b) When the region is rotated about the x-axis, each cross section is a washer with outer radius $R = 10 - x^2$ and inner radius $r = x^2 + 2$.

(c) The volume of the solid of revolution is

$$\pi \int_{-2}^{2} \left((10 - x^2)^2 - (x^2 + 2)^2 \right) dx = \pi \int_{-2}^{2} (96 - 24x^2)\, dx = \pi \left(96x - 8x^3 \right) \Big|_{-2}^{2} = 256\pi.$$

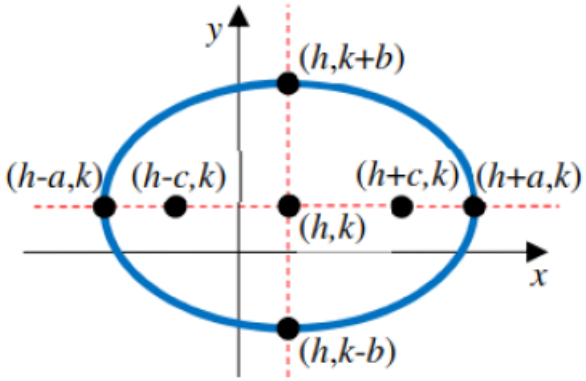
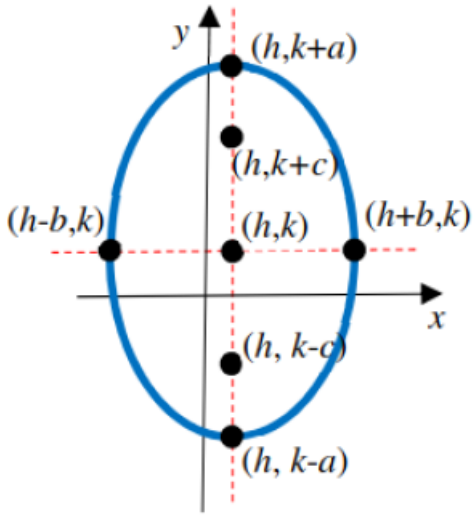
Conic Sections:



Circle Ellips Parabola Hyperbola

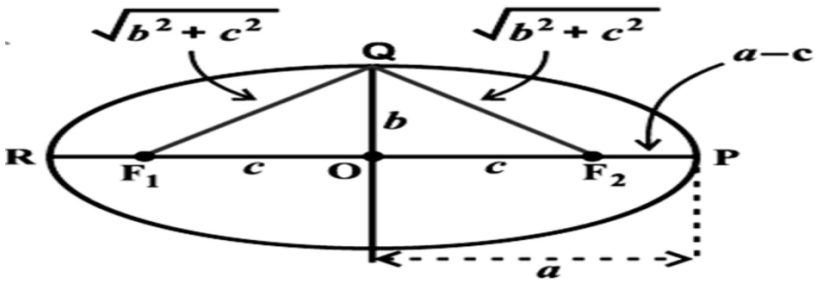


Ellipse:

Major Axis	Horizontal	Vertical
Standard Equation	$\frac{(x-h)^2}{a^2}+\frac{(y-k)^2}{b^2}=1$	$\frac{(x-h)^2}{b^2}+\frac{(y-k)^2}{a^2}=1$
Vertices	$(h \pm a, k)$	$(h, k \pm a)$
Minor Axis Endpoints	$(h, k \pm b)$	$(h \pm b, k)$
Foci	$(h \pm c, k)$ where $b^2 = a^2 - c^2$	$(h, k \pm c)$ where $b^2 = a^2 - c^2$
Graph		

Major axis is horizontal: center at (h, k)

Equation	$\frac{(x-h)^2}{a^2}+\frac{(y-k)^2}{b^2}=1\ (a>b)$
Center-to-focus distance	$c=\sqrt{a^2-b^2}$
Foci	$(h \pm c, k)$
Vertices	$(h \pm a, k)$
Endpoints of the minor axis	$(h, k \pm b)$



Ellipses with Center (h, k)

• **Standard equation**	$\frac{(x-h)^2}{a^2}+\frac{(y-k)^2}{b^2}=1$
• **Focal axis**	$y=k$
• **Foci**	$(h \pm c, k)$
• **Vertices**	$(h \pm a, k)$
• **Semimajor axis**	a
• **Semiminor axis**	b
• **Pythagorean relation**	$a^2=b^2+c^2$

120. For the ellipse

$$\frac{(x-3)^2}{100}+\frac{(y+2)^2}{36}=1$$

Find center, foci, vertices, end points of minor axis , length of major axis and length of minor axis.

Solution:

$$h=3 \qquad k=-2$$

$$a=10 \quad ,b=6 \quad \Rightarrow \quad c=\sqrt{100-36}=8$$

Center (h,k)	$(3,-2)$
Foci $(h \pm c, k)$	$(11,-2)\,,(-5,-2)$
Vertices $(h \pm a, k)$	$(13,-2)\,,(-7,-2)$
End points of minor axis $(h, k \pm b)$	$(3,4)\,,(3,-8)$
Length of major axis $2a$	20
Length of minor axis $2b$	12

121. Find the standard form of the equation for the ellipse whose major axis has endpoints $(-2, -1)$ and $(8, -1)$, and whose minor axis has length 8.

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1,$$

where the center (h, k) is the midpoint $(3, -1)$ of the major axis. The semimajor axis and semiminor axis are

$$a = \frac{8-(-2)}{2} = 5 \quad \text{and} \quad b = \frac{8}{2} = 4.$$

So the equation we seek is

$$\frac{(x-3)^2}{5^2} + \frac{(y-(-1))^2}{4^2} = 1,$$

$$\frac{(x-3)^2}{25} + \frac{(y+1)^2}{16} = 1.$$

122. ***Find the equation of the ellipse centered at $(3, -1)$ with a horizontal major axis of length 10 units and vertical minor axis of length 4 units.***

Solution. The given center $(3, -1) \Longrightarrow h = 3$ and $k = -1$. Horizontal major axis of length $10 \Longrightarrow 2a = 10 \Longrightarrow a = 5$. Vertical minor axis of length $4 \Longrightarrow 2b = 4 \Longrightarrow b = 2$. The equation of the ellipse is

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

$$\frac{(x-3)^2}{5^2} + \frac{(y+1)^2}{2^2} = 1$$

$$\frac{(x-3)^2}{25} + \frac{(y+1)^2}{4} = 1$$

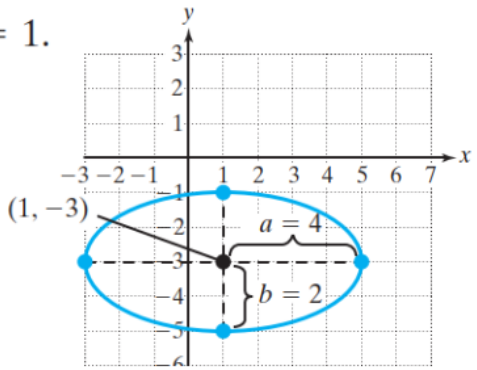
123. Graph the ellipse give by the equation. $\dfrac{(x-1)^2}{16}+\dfrac{(y+3)^2}{4}=1$

Solution:

Just as we would find the center of a circle, we see that the center of the ellipse is $(1, -3)$. Now we can use the values of a and b to help us plot four strategic points to define the curve.

The equation can be written as $\dfrac{(x-1)^2}{(4)^2}+\dfrac{(y+3)^2}{(2)^2}=1.$

From this, we have $a = 4$ and $b = 2$. To sketch the curve, locate the center at $(1, -3)$. Then move $a = 4$ units to the left and right of the center and plot two points. Similarly, move $b = 2$ units up and down from the center and plot two points.



124. Write the equation $2x^2+8y^2-20x+48y=-106$ in standard form and sketch a graph of the ellipse.

$$2x^2+8y^2-20x+48y=-106$$

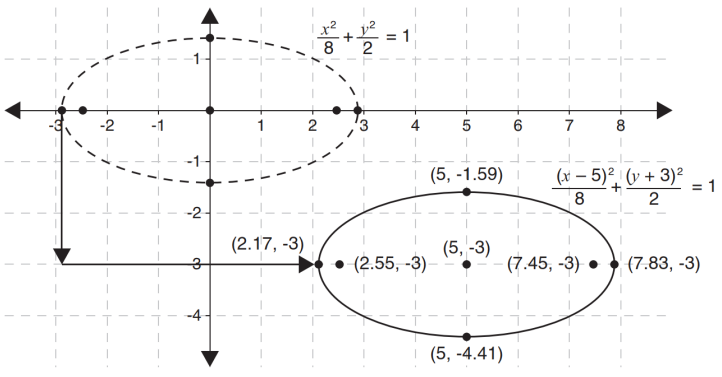
$$2(x^2-10x)+8(y^2+6y)=-106 \qquad \text{Regroup and factor.}$$

$$2(x^2-10x+25)+8(y^2+6y+9)=-106+122 \qquad 2\left(\frac{10}{2}\right)^2=50,\ 8\left(\frac{6}{2}\right)^2=72$$

$$2(x-5)^2+8(y+3)^2=16 \qquad \text{Factor.}$$

$$\frac{(x-5)^2}{8}+\frac{(y+3)^2}{2}=1 \qquad \text{Divide both sides by 16.}$$

Since the denominator of x^2 is larger, the ellipse has a horizontal major axis and the standard equation is $\dfrac{(x-h)^2}{a^2}+\dfrac{(y-k)^2}{b^2}=1.$
The center of the ellipse is (5, –3).



$a^2 = 8$ $\quad b^2 = 2$ $\quad c^2 = a^2 - b^2$

$a = \sqrt{8}$ $\quad b^2 = 2$ $\quad c^2 = 8 - 2$

$a \approx 2.83$ $\quad b \approx 1.41$ $\quad c^2 = 6$

$c \approx 2.45$

Foci: $(h - c, k), (h + c, k)$

$(5 - 2.45, -3), (5 + 2.45, -3)$

$(2.55, -3), (7.45, -3)$

Vertices: $(h - a, k), (h + a, k)$

$(5 - 2.83, -3), (5 + 2.83, -3)$

$(2.17, -3), (7.83, -3)$

Co-Vertices: $(h, k - b), (h, k + b)$

$(5, -3 - 1.41), (5, -3 + 1.41)$

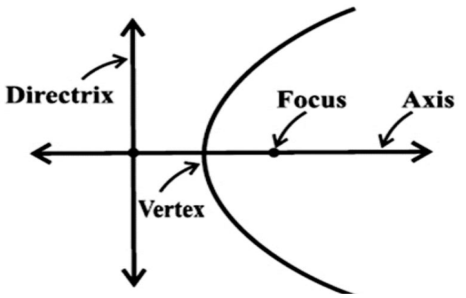
$(5, -4.41), (5, -1.59)$

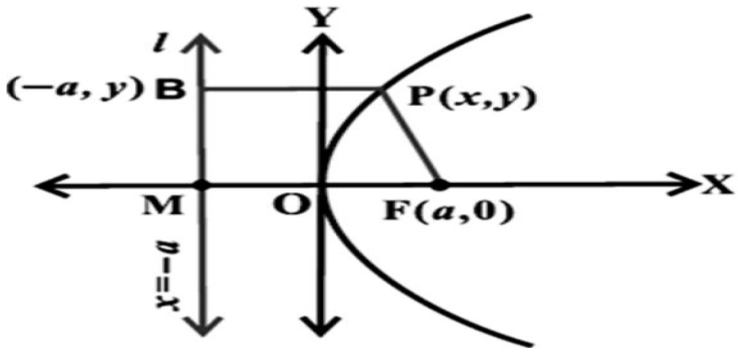
Notice in the previous example that you could also graph the ellipse by translating the graph of $\frac{x^2}{8} + \frac{y^2}{2} = 1$ down 3 and to the right 5.

You can also write the equation of an ellipse if you know the coordinates of the vertices and the co-vertices.

Parabola

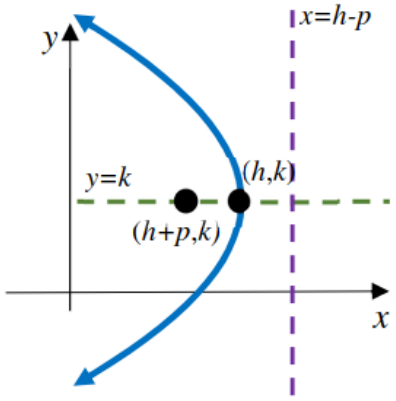
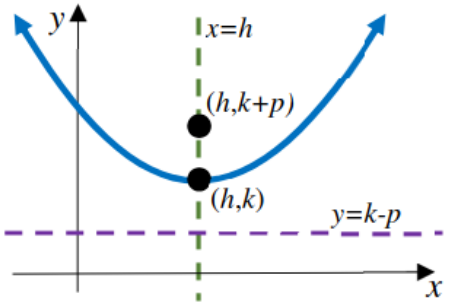
A line through the focus and perpendicular to the directrix is called the axis of the parabola. The point of intersection of parabola with the





Equation of a Parabola with Vertex at (*h*, *k*) in Standard Conic Form

The standard conic form of an equation of a parabola with vertex at the point (h,k) depends on whether the axis of symmetry is horizontal or vertical. The table below gives the standard equation, vertex, axis of symmetry, directrix, focus, and graph for each.

	Horizontal	Vertical
Standard Equation	$(y-k)^2 = 4p(x-h)$	$(x-h)^2 = 4p(y-k)$
Vertex	(h, k)	(h, k)
Axis of symmetry	$y = k$	$x = h$
Directrix	$x = h - p$	$y = k - p$
Focus	$(h + p, k)$	$(h, k + p)$
Graph	An example with $p < 0$ 	An example with $p > 0$ 

125 A parabola has its vertex at (1,5) and focus at (3,5). Find an equation for the parabola.

Since the vertex and focus lie on the line $y = 5$, that is our axis of symmetry.

The vertex (1,5) tells us $h = 1$ and $k = 5$.

Looking at the distance from the vertex to the focus, $p = 3 - 1 = 2$.

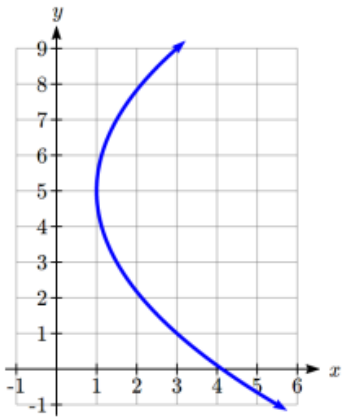
Substituting these values into the standard conic form of an equation for a horizontal parabola gives the equation

$$(y-5)^2 = 4(2)(x-1)$$

$$(y-5)^2 = 8(x-1)$$

Note this could also be rewritten by solving for x, resulting in

$$x = \frac{1}{8}(y-5)^2 + 1$$



126 Find the coordinates of the focus, the equation of axis and the equation of directrix for the parabola $y^2 = -8x$.

Solution:

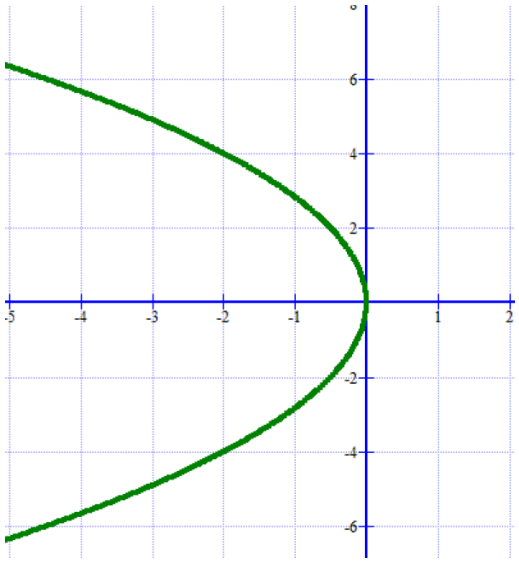
$$-4a = -8 \quad \Rightarrow \quad a = 2$$

Vertex (0,0)

Focus (-2,0)

Directrix $x = 2$

Axis $y = 0$



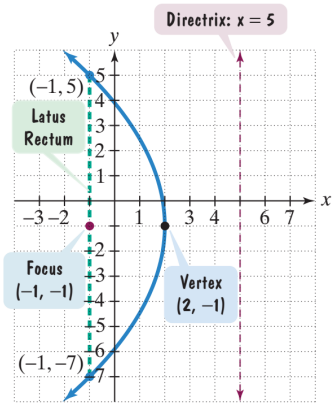


Figure 9.38 The graph of $y^2 + 2y + 12x - 23 = 0$, or $(y + 1)^2 = -12(x - 2)$

127 Find the vertex, focus, and directrix of the parabola given by

$$y^2 + 2y + 12x - 23 = 0.$$

$$y^2 + 2y + 12x - 23 = 0$$ This is the given equation.

$$y^2 + 2y = -12x + 23$$ Isolate the terms involving y.

$$y^2 + 2y + 1 = -12x + 23 + 1$$ Complete the square by adding the square of half the coefficient of y.

$$(y + 1)^2 = -12x + 24$$ Factor.

To express the equation $(y + 1)^2 = -12x + 24$ in the standard form $(y - k)^2 = 4p(x - h)$, we factor out -12 on the right. The standard form of the parabola's equation is

$(y + 1)^2 = -12(x - 2)$ to identify the vertex, (h, k), and the value for p needed to locate the fo

$[y - (-1)]^2 = -12($

This is $(y - k)^2$, with $k = -1$.

This is $4p$.

quared

parabola's

Focus: $(h + p, k) = (2 + (-3), -1) = (-1, -1)$

The vertex, (h, k), is $(2, -1)$.

The focus is 3 units to the left of the vertex, $(2, -1)$.

Directrix: $x = h - p$

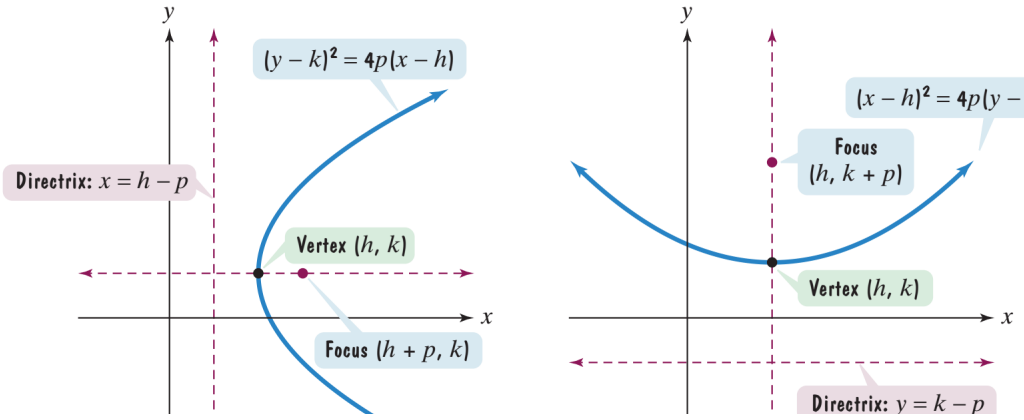
$x = 2 - (-3) = 5$

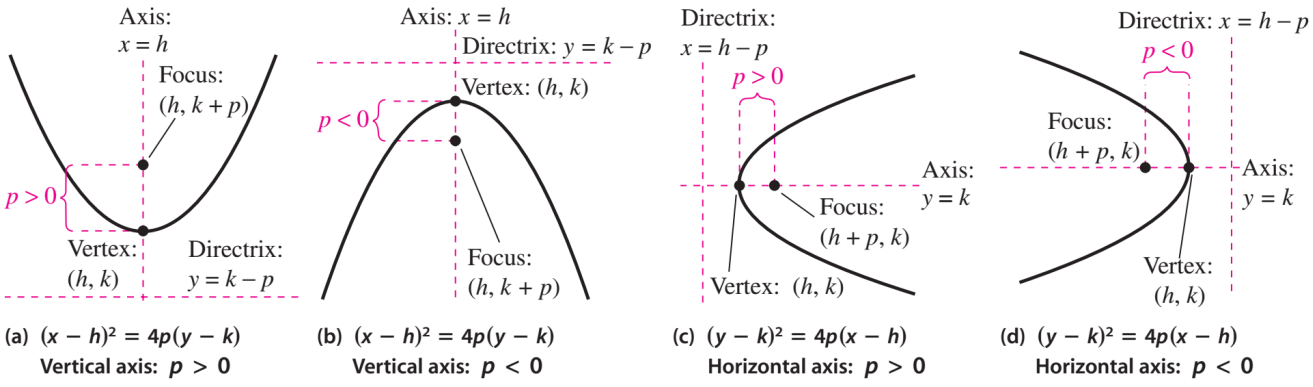
The directrix is 3 units to the right of the vertex, $(2, -1)$.

We see that $h = 2$ and $k = -1$. Thus, the vertex of the parabola is $(h, k) = (2, -1)$. Because $4p = -12$, $p = -3$. Based on the standard form of the equation, the axis of symmetry is horizontal. With a negative value for p and a horizontal axis of symmetry, the parabola opens to the left. Because $p = -3$, the focus is located 3 units to the left of the vertex, $(2, -1)$. Likewise, the directrix is located 3 units to the right of the vertex.

Standard Forms of Equations of Parabolas with Vertex at (h, k)

Equation	Vertex	Axis of Symmetry	Focus	Directrix	Description
$(y - k)^2 = 4p(x - h)$	(h, k)	Horizontal	$(h + p, k)$	$x = h - p$	If $p > 0$, opens to the right. If $p < 0$, opens to the left.
$(x - h)^2 = 4p(y - k)$	(h, k)	Vertical	$(h, k + p)$	$y = k - p$	If $p > 0$, opens upward. If $p < 0$, opens downward.





(a) $(x - h)^2 = 4p(y - k)$ Vertical axis: $p > 0$

(b) $(x - h)^2 = 4p(y - k)$ Vertical axis: $p < 0$

(c) $(y - k)^2 = 4p(x - h)$ Horizontal axis: $p > 0$

(d) $(y - k)^2 = 4p(x - h)$ Horizontal axis: $p < 0$

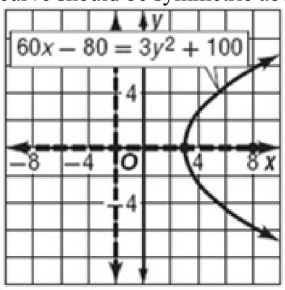
128 Find the vertex, focus, and directrix of the parabola given by $60x - 80 = 3y^2 + 100$

$$60x - 80 = 3y^2 + 100$$
$$60x - 180 = 3y^2$$
$$20x - 60 = y^2$$
$$20(x - 3) = y^2$$
$$y^2 = 20(x - 3)$$

Because the y–term is squared and $p = 5$, the graph opens to the right. Use the standard form to determine the characteristics of the parabola.

vertex: $(h, k) = (3, 0)$
focus: $(h + p, k) = (8, 0)$
directrix: $x = h - p$ or -2
axis of symmetry: $y = k$ or 0

Graph the vertex, focus, axis, and directrix of the parabola. Then make a table of values to graph the curve. The curve should be symmetric about the axis of symmetry.



129 Consider the equation $y^2 + 4y + 8x = 4$. Put this equation into standard form and identify the vertex, focus, directrix, and graph.

$y^2 + 4y + 8x = 4$

$y^2 + 4y = -8x + 4$

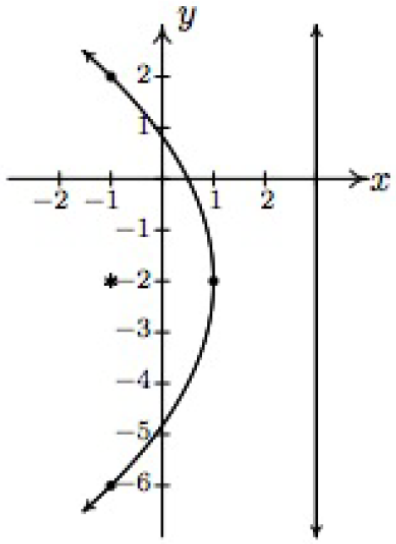
$y^2 + 4y + 4 = -8x + 4 + 4$

$(y + 2)^2 = -8x + 8$

$(y + 2)^2 = -8(x + 1)$

complete the square in y only

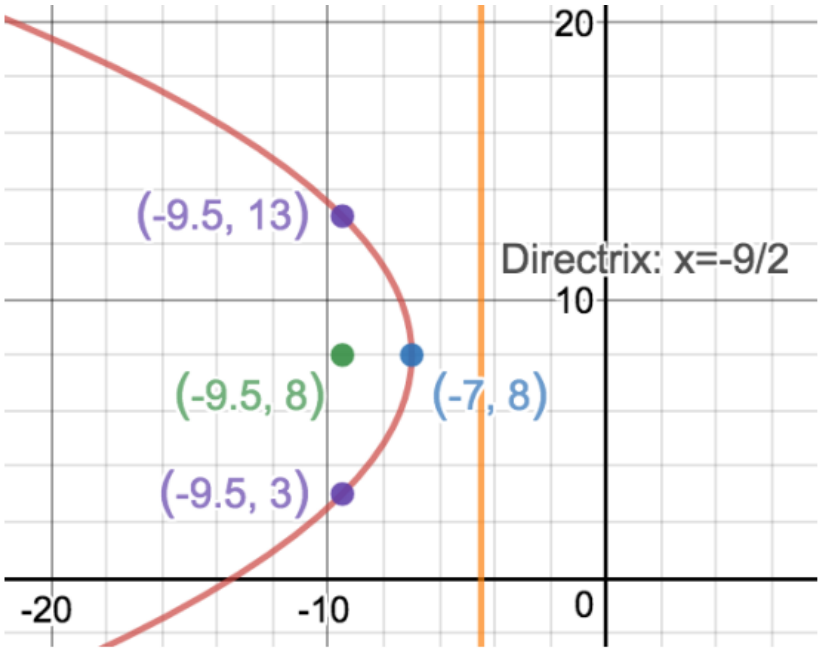
factor



Vertex: $(1, -2)$
Focus: $(-1, -2)$
Directrix: $x = 3$

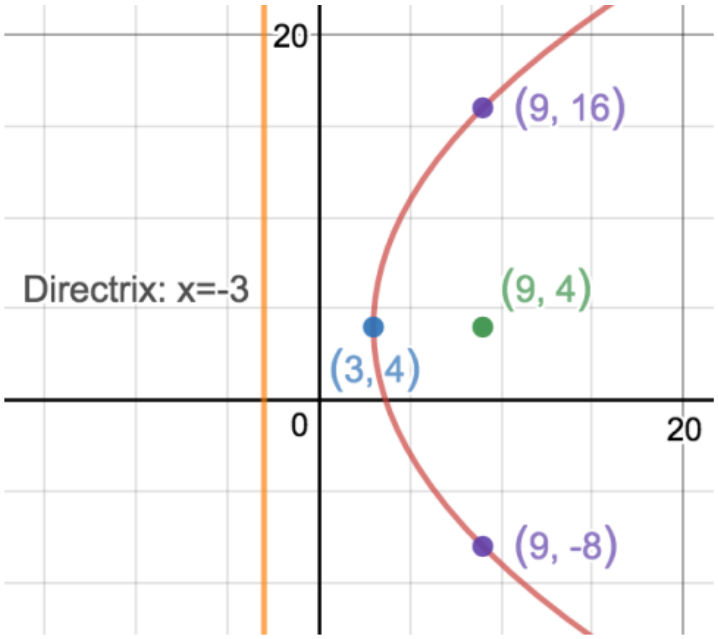
130 sketch the graph of the given parabola. Find the vertex, focus, and directrix.

$$(y-8)^2 = -10(x+7)$$



131 sketch the graph of the given parabola. Find the vertex, focus, and directrix.

$$(y-4)^2 = 24(x-3)$$



132 $y^2-6y-36x+117=0$

$y^2-6y+\underline{3^2}=36x-117+\underline{9}$

$(y-3)^2=36x-108$

$(y-3)^2=36(x-3)$

Vertex: (3,3)

Focus: (12,3)

Directrix: x=-6

133 Write the given equation in the standard form. Determine the coordinates of the center, vertices and foci.

$2x^2+3y^2+8x-6y+5=0$

1. Re-write the equation so the x-terms and the y-terms are together; move the number to the right hand side

$(2x^2+8x)+(3y^2-6y)=-5$

2. From the x- group, factor out the coefficient of x^2; from the y-group- factor the coefficient of y^2

$2(x^2+4x)+3(y^2-2y)=-5$

3. Complete the square in each group; add appropriate numbers to the right hand side

$2(x^2+4x+4)+3(y^2-2y+1)=-5+2\cdot4+3\cdot1$

4. Write each group as a perfect square; add numbers on the right

$2(x+2)^2+3(y-1)^2=6$

5. Divide each term by the number on the right hand side (6) and simplify

$$\frac{2(x+2)^2}{6}+\frac{3(y-1)^2}{6}=\frac{6}{6}$$

$$\frac{(x+2)^2}{3}+\frac{(y-1)^2}{2}=1$$

This is the standard equation of the ellipse. ***Center*** *is at (-2,1).*

Hence, this ellipse is obtained by shifting the ellipse $\frac{x^2}{3}+\frac{y^2}{2}=1$ *two units to the left and one unit up. The numbers in the denominators are 3 and 2 and since 3 > 2, the vertices of* $\frac{x^2}{3}+\frac{y^2}{2}=1$ *are on the x-axis and* $a^2=3$ *and* $b^2=2$*, or* $a=\sqrt{3}$ *and* $b=\sqrt{2}$ *(remember that* $a^2>b^2$*) . Therefore, the* ***vertices*** *of this ellipse are at* $(\pm\sqrt{3},0)$*. Moreover, since* $c^2=a^2-b^2=3-2=1$*, c= 1, the foci are at (±1, 0).*

Therefore, for $\frac{(x+2)^2}{3}+\frac{(y-1)^2}{2}=1$*, the vertices are* $(\pm\sqrt{3}-2,1)$ *and the foci are (±1-2, 1) or (-3,1) and (1,-1).*